Approximation and String Algorithms

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These are the lecture notes for the M1 course Approximation and String Algorithms at ENS Paris-Saclay in the academic year 2026–2027. Good references for the material covered here include Williamson and Shmoys [1] for approximation algorithms and Lothaire [29] for the combinatorics on words underlying the string algorithms. The website for the course is at `https://algo.biodis.co/`.

This document will inevitably contain some errors. If you find any, I will be grateful and happy to hear from you. You can reach me by email at thomas@thomasnowak.net.

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Lecture 1: Approximation Algorithms I

Many fundamental optimization problems are NP-hard, meaning that finding exact optimal solutions is computationally infeasible for large instances. Approximation algorithms provide a principled approach to finding near-optimal solutions efficiently, with provable guarantees on solution quality.

1.1 Definitions

An optimization problem asks us to find a feasible solution that minimizes (or maximizes) some objective function. For a minimization problem, let OPT denote the optimal solution value and ALG the value produced by algorithm A.

Definition 1.1

Algorithm A is an α-approximation algorithm if

$$\text{ALG} \leq \alpha \cdot \text{OPT}$$

for all instances of the problem, where $\alpha \geq 1$ is called the approximation ratio.

For maximization problems, we require $\text{ALG} \geq \text{OPT}/\alpha$. The closer α is to 1, the better the approximation guarantee. An approximation ratio of $\alpha = 2$ means our solution is at most twice as expensive as optimal.

The landscape of approximable problems is characterized by several complexity classes. The class APX contains problems that admit constant-factor polynomial-time approximation algorithms. These are the "nice" NP-hard problems: hard to solve exactly, but not impossibly hard to approximate. A Polynomial-Time Approximation Scheme (PTAS) is more refined: for any $\varepsilon > 0$, there exists a $(1+\varepsilon)$-approximation algorithm running in time polynomial in n (though possibly exponential or worse in $1/\varepsilon$). For example, running time $O(n^{1/\varepsilon})$ is allowed, which becomes impractical for small ε. The strongest guarantee is a Fully Polynomial-Time Approximation Scheme (FPTAS), which runs in time polynomial in both n and $1/\varepsilon$, such as $O(n^2/\varepsilon)$.

Not all NP-hard problems admit good approximations. Some problems are provably hard to approximate within any reasonable factor unless P = NP, while others admit PTAS or even FPTAS.

1.2 Vertex Cover

The vertex cover problem serves as our first case study. Given an undirected graph $G = (V, E)$, a vertex cover is a subset $C \subseteq V$ such that every edge has at least one endpoint in C. We want to find a vertex cover of minimum size.

A natural greedy approach is to repeatedly select the vertex of maximum degree, add it to the cover, and remove all incident edges. Let us analyze both the performance and limitations of this approach.

Theorem 1.1. The greedy maximum degree algorithm achieves an approximation ratio of $O(\log n)$ for vertex cover.

Proof. Let C be the cover produced by the greedy algorithm. We will track the number of edges remaining after each iteration. Let $E_0 = E$ and let E_i denote the set of edges remaining after the i-th iteration, when we have selected i vertices for the cover.

In iteration $i+1$, we select a vertex v of maximum degree Δ_i in the graph (V, E_i). This vertex covers Δ_i edges, so $|E_{i+1}| = |E_i| - \Delta_i$.

Now consider the optimal solution, which uses at most OPT vertices to cover all edges. In particular, these OPT vertices must cover the $|E_i|$ edges that remain in E_i. By the pigeonhole principle, at least one vertex in the optimal solution must cover at least $|E_i|/\mathrm{OPT}$ of these edges.

Since the greedy algorithm selects the vertex of maximum degree, we have:

$$\Delta_i \geq \frac{|E_i|}{\mathrm{OPT}}$$

Therefore:

$$|E_{i+1}| = |E_i| - \Delta_i \leq |E_i| - \frac{|E_i|}{\mathrm{OPT}} = |E_i| \left(1 - \frac{1}{\mathrm{OPT}}\right)$$

Unrolling this recurrence from $i = 0$ to $i = t - 1$ where t is the number of iterations:

$$|E_t| \leq |E_0| \left(1 - \frac{1}{\mathrm{OPT}}\right)^t$$

The algorithm terminates when $|E_t| = 0$, so we need:

$$|E_0| \left(1 - \frac{1}{\mathrm{OPT}}\right)^t < 1$$

Using the inequality $(1 - x) \leq e^{-x}$ for $x > 0$:

$$|E_0| \left(1 - \frac{1}{\mathrm{OPT}}\right)^t \leq |E_0|\, e^{-t/\mathrm{OPT}}$$

The right-hand side is less than 1 when $t > \mathrm{OPT} \cdot \log|E_0|$. Therefore the algorithm terminates with at most $\lceil \mathrm{OPT} \cdot \log|E_0| \rceil$ vertices. Since $|E_0| \leq \binom{n}{2} < n^2$, we have $|C| \leq \mathrm{OPT} \cdot \log n^2 + 1 = O(\mathrm{OPT} \cdot \log n)$. □

This seems promising: a logarithmic approximation ratio. However, the greedy approach can perform poorly on certain graph families.

Theorem 1.2. There exists a family of graphs on which the greedy maximum degree algorithm produces a cover of size $\Omega(\mathrm{OPT} \cdot \log n)$.

Proof. Consider the family of bipartite graphs $G_r = (L \cup R, E)$ constructed as follows. Let $L = \{\ell_1, \ldots, \ell_r\}$ and partition $R = R_1 \cup R_2 \cup \cdots \cup R_r$ where $|R_i| = \lfloor r/i \rfloor$. We connect each vertex in R_i to exactly i distinct vertices in L, subject to the constraint that every vertex in L has at most one neighbor in R_i. This is possible because R_i needs $i \cdot \lfloor r/i \rfloor \leq r$ distinct endpoints in L.

Concretely, label the vertices in R_i as $v_1^{(i)}, \ldots, v_{\lfloor r/i \rfloor}^{(i)}$ and connect $v_j^{(i)}$ to each of the i vertices $\ell_{(j-1)i+1}, \ell_{(j-1)i+2}, \ldots, \ell_{ji}$. This way, within each group R_i, the neighborhoods partition (a subset of) L into blocks of size i, so no vertex in L has more than one neighbor in R_i. As a result, every vertex in R_i has degree exactly i, and every vertex $\ell_j \in L$ has degree $|\{i : 1 \leq i \leq r,\ \ell_j \text{ has a neighbor in } R_i\}|$.

See Figure 1.1 for an illustration with $r = 4$.

The optimal solution is $\mathrm{OPT} = r$ (take all vertices in L, which covers every edge). However, the greedy algorithm can be forced to select all vertices in R. The maximum degree in G_r is r, attained by the single vertex of R_r and by some vertices of L as well, for instance ℓ_1, which has a neighbor in every group; we let greedy break ties in favor of R, so it selects the vertex of R_r first. In general, once the groups $R_r, \ldots, R_{i+1}$ have been removed, every vertex of R_i still has degree i, since only edges at removed vertices have disappeared, whereas every vertex of L has lost its neighbors in the removed groups and has degree at most i. With ties broken in favor of R, greedy therefore picks all of R_i before touching L, for $i = r, r-1, \ldots, 1$.

The total number of vertices selected is:

$$|R| = \sum_{i=1}^{r} \lfloor r/i \rfloor \approx r \sum_{i=1}^{r} \frac{1}{i} = r \cdot H_r \approx r \log r = \Omega(\mathrm{OPT} \cdot \log r)$$

This concludes the proof. □

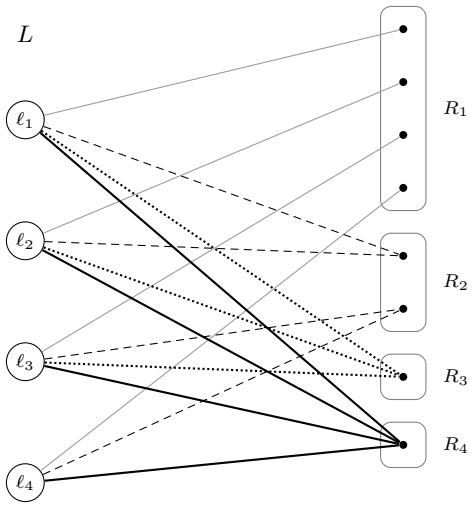


Figure 1.1: The graph G_4 with $r = 4$. Each vertex in R_i has degree exactly i. Edge styles distinguish groups: R_1 thin gray, R_2 dashed, R_3 dotted, R_4 thick solid.

This shows the greedy algorithm's approximation ratio is tight at $\Theta(\log n)$. Can we achieve a *constant-factor* approximation? Rather than guessing an algorithm and analyzing it after the fact, let us think about what a proof of a c-approximation would have to look like, and let the proof structure guide us to an algorithm. Suppose an algorithm produces a cover C and we claim $|C| \leq c \cdot \mathrm{OPT}$. Since computing OPT is NP-hard, the proof cannot simply compare $|C|$ to the optimal value. Instead, the proof must exhibit some efficiently verifiable *certificate* that OPT is large enough, that is, that $\mathrm{OPT} \geq |C|/c$.

What kind of certificate can witness that every vertex cover is large? Each edge imposes a *demand*: at least one of its endpoints must be in the cover. A single edge certifies $\mathrm{OPT} \geq 1$; two edges certify $\mathrm{OPT} \geq 2$—but only if they don't share an endpoint, since otherwise a single vertex could satisfy both demands simultaneously. More generally, to certify $\mathrm{OPT} \geq k$, we need k edges whose demands are independent, meaning no two of them can be satisfied by the same vertex. This happens exactly when the edges are *pairwise disjoint*: they share no endpoints.

So a c-approximation proof needs a set M of pairwise disjoint edges with $|C| \leq c \cdot |M|$. What if the

algorithm itself builds M as a byproduct? The simplest idea: take both endpoints of every edge in M as the cover, giving $|C| = 2|M|$ and thus $c = 2$. But is this actually a valid vertex cover? An edge $e \notin M$ could escape coverage if neither of its endpoints belongs to C. This cannot happen if M is *maximal*: if every edge in E shares an endpoint with some edge of M, then every edge is covered. This leads us to the following algorithm.

Algorithm:

1. Initialize $C \leftarrow \emptyset$ and $E' \leftarrow E$.
2. While $E' \neq \emptyset$:
   (a) Pick any edge $\{u, v\} \in E'$.
   (b) Add both u and v to C.
   (c) Remove all edges incident to u or v from E'.
3. Return C.

Note that this procedure computes a maximal matching M (the set of picked edges) and returns both endpoints of every edge in M.

Theorem 1.3. The edge-picking algorithm is a 2-approximation for vertex cover.

Proof. Let M denote the set of edges selected by the algorithm. By construction, M is a maximal matching: edges in M are pairwise disjoint, and no remaining edge can be added because all edges incident to selected vertices have been removed. Since we add both endpoints of each selected edge, $|C| = 2|M|$.

By maximality, every edge in E is incident to at least one vertex in C, so C is a valid vertex cover. Since the edges in M are pairwise disjoint, any vertex cover must include at least one distinct endpoint per edge, so $\mathrm{OPT} \geq |M|$. Therefore:

$$|C| = 2|M| \leq 2 \cdot \mathrm{OPT}$$

□

The 2-approximation for vertex cover has been known since the 1970s, and despite decades of effort, no polynomial-time algorithm with a better constant factor has been found. It is natural to ask whether we can do better than the 2-approximation.

On the hardness side, Dinur and Safra [2] proved that it is NP-hard to approximate vertex cover within a factor of 1.3606. This means that unless P = NP, no polynomial-time algorithm can guarantee a solution strictly better than $1.3606 \cdot \mathrm{OPT}$. Even stronger hardness results are known under the Unique Games Conjecture (UGC), introduced by Khot [4]. The UGC posits that a certain type of constraint satisfaction problem, where each constraint uniquely determines one variable given the other, is hard to approximate. Under this assumption, Khot and Regev [5] showed that vertex cover cannot be approximated within any factor $2 - \varepsilon$ for constant $\varepsilon > 0$. If the UGC is true, then our simple 2-approximation algorithm is essentially optimal. The gap between the 2-approximation upper bound and the current best hardness lower bound of 1.3606 (without assuming UGC) remains one of the major open problems in approximation algorithms.

1.3 Set Cover

The set cover problem generalizes vertex cover and many other covering problems. We are given a universe U with $|U| = n$ and a collection of subsets $\mathcal{S} = \{S_1, S_2, \ldots, S_m\}$ where $S_i \subseteq U$. The goal is to find a minimum cardinality subcollection $\mathcal{C} \subseteq \mathcal{S}$ such that the selected sets cover all elements, that is, $\bigcup_{S \in \mathcal{C}} S = U$.

Lemma 1.1. Vertex cover is a special case of set cover.

Proof. Consider a graph $G = (V, E)$. Let $U = E$ be the universe (the edges), and for each vertex $v \in V$, let $S_v = \{e \in E : v \in e\}$ be the set of edges incident to v. Then a vertex cover corresponds exactly to a set cover: we select vertices (sets) such that every edge (element) is covered.

Formally, $C \subseteq V$ is a vertex cover if and only if $\{S_v : v \in C\}$ is a set cover of $U = E$. Since $|C| = |\{S_v : v \in C\}|$, the optimal solutions have the same size. □

The natural greedy algorithm makes the locally optimal choice at each step: repeatedly select the set that covers the most uncovered elements.

Greedy Set Cover Algorithm:

1. Initialize $\mathcal{C} \leftarrow \emptyset$ and $R \leftarrow U$.
2. While $R \neq \emptyset$:
   (a) Select $S_i \in \mathcal{S}$ that maximizes $|S_i \cap R|$.
   (b) Add S_i to $\mathcal{C}$.
   (c) Update $R \leftarrow R \setminus S_i$.
3. Return $\mathcal{C}$.

To analyze this algorithm, we use a charging argument that assigns costs to elements as they are covered. This is a powerful technique: instead of directly bounding the number of sets, we assign a cost to each element and show the total cost is bounded. The beauty of this approach is that it reduces a combinatorial counting problem to an accounting problem.

The analysis uses the following bound on the harmonic numbers $H_n = \sum_{i=1}^{n} \frac{1}{i}$. It corresponds to a special case of the Euler–Maclaurin summation formula. Its basic message is that oftentimes it is possible to exchange an integral for a sum.

Lemma 1.2. For all $n \in \mathbb{N}$, we have $\log(n+1) \leq H_n \leq 1 + \log n$.

Proof. For the upper bound, we calculate:

$$\begin{aligned} H_n &= \sum_{i=1}^{n} \frac{1}{i} = 1 + \sum_{i=2}^{n} \frac{1}{i} \int_{i-1}^{i} 1 \, dx = 1 + \sum_{i=2}^{n} \int_{i-1}^{i} \frac{1}{i} \, dx \leq 1 + \sum_{i=2}^{n} \int_{i-1}^{i} \frac{1}{x} \, dx \\ &= 1 + \int_{1}^{n} \frac{1}{x} \, dx = 1 + \log n \end{aligned}$$

For the lower bound, we note:

$$\begin{aligned} H_n &= \sum_{i=1}^{n} \frac{1}{i} = \sum_{i=1}^{n} \frac{1}{i} \int_{i}^{i+1} 1 \, dx = \sum_{i=1}^{n} \int_{i}^{i+1} \frac{1}{i} \, dx \geq \sum_{i=1}^{n} \int_{i}^{i+1} \frac{1}{x} \, dx \\ &= \int_{1}^{n+1} \frac{1}{x} \, dx = \log(n+1) \end{aligned}$$

This concludes the proof. □

Theorem 1.4. The greedy algorithm for set cover is an $O(\log n)$-approximation.

Proof. We assign a cost to each element when it is covered by the greedy algorithm. Consider the iteration when the greedy algorithm selects a set S covering t new elements. We charge a cost of $1/t$ to each of these newly covered elements, so the total cost assigned in this iteration is exactly 1. Since the algorithm selects exactly $|\mathcal{C}|$ sets, and each set is "charged" a total cost of 1, the total cost equals $|\mathcal{C}|$.

Let $k = |R|$ denote the number of uncovered elements remaining before this iteration. Since an optimal solution must cover all n elements using at most OPT sets, by the pigeonhole principle at least one of the sets in the optimal solution must cover at least k/OPT of the remaining uncovered elements. The greedy algorithm selects the set covering the most uncovered elements, so it covers at least k/OPT elements, that is, $t \geq k/\mathrm{OPT}$.

Therefore, the cost charged per element in this iteration is:

$$\frac{1}{t} \leq \frac{\mathrm{OPT}}{k}$$

Now consider the perspective of a single element. Think about the element that is the j-th element to be covered, counting backwards from the end. That is, the last element covered has $j = 1$, the second-to-last has $j = 2$, and so on. When this element is covered, there are at least j elements still uncovered (including itself), so $k \geq j$. Thus, the cost assigned to this element is at most:

$$\frac{1}{t} \leq \frac{\mathrm{OPT}}{k} \leq \frac{\mathrm{OPT}}{j}$$

Summing over all n elements, the total cost assigned is:

$$\mathrm{Cost} \leq \sum_{j=1}^{n} \frac{\mathrm{OPT}}{j} = \mathrm{OPT} \sum_{j=1}^{n} \frac{1}{j} = \mathrm{OPT} \cdot H_n$$

Since each set selected by the greedy algorithm is assigned a total cost of 1 (distributed among the elements it covers), and the algorithm selects $|\mathcal{C}|$ sets, we have:

$$|\mathcal{C}| = \mathrm{Cost} \leq \mathrm{OPT} \cdot H_n$$

By Lemma 1.2, we have $H_n \leq 1 + \log n$, completing the proof. □

The charging argument is a fundamental technique in approximation algorithms. We will see it again in other contexts. The key idea is always the same: assign costs to objects (here, elements), argue that greedy makes good local decisions about these costs, and sum up to get a global bound.

Just as with vertex cover, it is natural to ask whether we can improve upon the logarithmic approximation ratio for set cover. The answer is largely negative. Feige [3] proved that set cover cannot be approximated within a factor of $(1 - \varepsilon) \log n$ for any $\varepsilon > 0$, unless NP has slightly unusual complexity (specifically, unless NP $\subseteq$ DTIME($n^{O(\log \log n)}$)). This shows that the greedy algorithm's $O(\log n)$-approximation is essentially optimal.

Two natural structural parameters control how far we can improve upon the $O(\log n)$ ratio for restricted instances: the *size* of the sets and the *frequency* of the elements.

If every set has size at most K, the greedy analysis tightens immediately. Keep the charging of Theorem 1.4, but add up the charges set by set instead of element by element. Fix a set S^* of an optimal cover and list its elements $f_1, \ldots, f_s$ in the order in which greedy covers them, with $s \leq K$. Just before f_i is covered, the elements $f_i, \ldots, f_s$ are all uncovered, so S^* alone would cover at least $s - i + 1$ new elements,

and greedy's choice covers at least as many: f_i is charged at most $1/(s-i+1)$. The elements of S^* therefore receive a total charge of at most $\sum_{i=1}^{s} 1/(s-i+1) = H_s \leq H_K$. Every element lies in some set of the optimal cover, so the total charge, which is the number of sets picked by greedy, is at most $\text{OPT} \cdot H_K$. The greedy algorithm is thus an H_K-approximation, which is $O(\log K)$; an improvement when $K \ll n$, and for $K = n$ a second proof of Theorem 1.4.

A stronger improvement comes from bounding the frequency. Define the *frequency* of an element as the number of sets containing it. If every element has frequency at most f, we can obtain an f-approximation by a simple rounding argument. Consider any optimal solution $\mathcal{C}^*$ with $|\mathcal{C}^*| = \text{OPT}$. For each element $e \in U$, assign it a weight of $1/f$. The total weight assigned to elements of any single set S_i is $|S_i|/f$. The total weight across all elements is n/f. On the other hand, we have $\sum_{S \in \mathcal{C}^*} |S| \geq n$. Now consider the algorithm that simply selects every set containing at least one uncovered element, greedily avoiding redundant sets: pick any uncovered element e; it belongs to at most f sets, and at least one of these sets is in $\mathcal{C}^*$. This means that any *maximal* sub-collection that covers U uses at most $f \cdot \text{OPT}$ sets, since each set in $\mathcal{C}^*$ "accounts for" at most f sets in our solution.

This explains why vertex cover admits a 2-approximation. Vertex cover is a special case of set cover (Lemma 1.1) where each element (edge) has frequency exactly 2: every edge belongs to exactly the two sets corresponding to its endpoints. The maximal matching algorithm from Theorem 1.3 is precisely an instantiation of the frequency-based argument with $f = 2$. Note also that the greedy maximum degree algorithm for vertex cover is the greedy set cover algorithm applied to this instance, so Theorem 1.4 gives an alternative $O(\log n)$ proof of Theorem 1.1.

1.4 Subset Sum

The vertex cover and set cover problems we've seen so far achieve constant-factor or logarithmic approximations. Can we do better? For some problems, the answer is yes: we can achieve arbitrarily good approximations.

The subset sum problem provides our first example of a PTAS and an FPTAS. Given positive integers $a_1, \ldots, a_n$ and a target T, we want to find a subset $S \subseteq \{1, \ldots, n\}$ such that $\sum_{i \in S} a_i$ is as close to T as possible without exceeding it.

The standard dynamic programming approach maintains a table $\text{DP}[i][s]$ which is true if we can achieve sum exactly s using a subset of the first i items. The recurrence is:

$$\text{DP}[i][s] = \text{DP}[i-1][s] \vee \text{DP}[i-1][s-a_i]$$

This runs in time $O(nT)$, which is pseudo-polynomial: it's polynomial in the value of T but exponential in the number of bits needed to represent T. The idea is to reduce the number of distinct sums we track by "trimming" the list of reachable sums. For any $\delta > 0$, we say two values s and s' are δ-close if $s \leq s' \leq (1+\delta)s$. When maintaining our list of reachable sums, whenever we have two values that are δ-close, we keep only the smaller one. Set $\delta = \varepsilon/(2n)$. We maintain a list L_i of reachable sums using items from $\{a_1, \ldots, a_i\}$, but we keep L_i trimmed so that no two values are δ-close.

Algorithm:

1. Initialize $L_0 \leftarrow \{0\}$.
2. For $i \leftarrow 1$ to n:
   (a) Compute $L_i' \leftarrow L_{i-1} \cup \{s + a_i : s \in L_{i-1}\}$.
   (b) Trim L_i' to obtain L_i:
   - Sort L_i' in increasing order.
   - Initialize $L_i \leftarrow \emptyset$ and last $\leftarrow 0$.
   - For each $s \in L_i'$ in sorted order:

- If $s \le T$, and L_i is empty or $s > (1+\delta) \cdot \text{last}$, add s to L_i and set last $\leftarrow s$.

3. Return the largest value in L_n.

Theorem 1.5. For any $\varepsilon > 0$, the above algorithm is a $(1+\varepsilon)$-approximation for subset sum, running in time $O(n^2 \log^2(nT)/\varepsilon)$.

Proof. We first bound the approximation error. Let OPT_i denote the maximum sum achievable using items from $\{a_1, \ldots, a_i\}$ that does not exceed T. We prove by induction that L_i contains a value y_i such that:

$$y_i \ge \frac{\text{OPT}_i}{(1+\delta)^i}$$

In the base case $i = 0$, we have $L_0 = \{0\}$ and $\text{OPT}_0 = 0$, so the claim holds. Now assume the claim holds for $i-1$. By the inductive hypothesis, there exists $y_{i-1} \in L_{i-1}$ with $y_{i-1} \ge \text{OPT}_{i-1}/(1+\delta)^{i-1}$. If $\text{OPT}_i = \text{OPT}_{i-1}$ (item a_i is not used in the optimal solution), then y_{i-1} is added to L'_i. During trimming, either y_{i-1} itself survives in L_i, or it is replaced by a value $z \le y_{i-1}$ with $z \ge y_{i-1}/(1+\delta)$. In either case, there exists $y_i \in L_i$ with $y_i \ge y_{i-1}/(1+\delta) \ge \text{OPT}_i/(1+\delta)^i$. Otherwise, $\text{OPT}_i = \text{OPT}_{i-1} + a_i$ for some optimal subset that includes a_i. Let OPT'_{i-1} be the sum of the optimal subset of $\{a_1, \ldots, a_{i-1}\}$ that is used in this optimal solution for OPT_i. Then $\text{OPT}_i = \text{OPT}'_{i-1} + a_i$. By the inductive hypothesis, there exists $y'_{i-1} \in L_{i-1}$ satisfying $y'_{i-1} \ge \text{OPT}'_{i-1}/(1+\delta)^{i-1}$. The value $y'_{i-1} + a_i$ is computed and added to L'_i. During trimming, either $y'_{i-1} + a_i$ itself survives, or it is removed because there is a smaller value z within a factor $(1+\delta)$ of it. In either case, there exists $y_i \in L_i$ with:

$$y_i \ge \frac{y'_{i-1} + a_i}{1+\delta} \ge \frac{\text{OPT}'_{i-1}/(1+\delta)^{i-1} + a_i}{1+\delta} \ge \frac{\text{OPT}_i}{(1+\delta)^i}$$

Therefore, the largest value in L_n is at least:

$$\frac{\text{OPT}}{(1+\delta)^n} = \frac{\text{OPT}}{(1+\varepsilon/(2n))^n}$$

Using $(1+x)^n \le e^{nx}$ for $x > 0$:

$$(1+\varepsilon/(2n))^n \le e^{\varepsilon/2}$$

For $\varepsilon \le 1$, we have $\varepsilon/2 \le \log(1+\varepsilon)$, so $e^{\varepsilon/2} \le 1+\varepsilon$. Hence, the algorithm returns a value at least $\text{OPT}/(1+\varepsilon)$. Since subset sum is a maximization problem, this means $\text{ALG} \ge \text{OPT}/(1+\varepsilon)$, so the algorithm is a $(1+\varepsilon)$-approximation.

We next bound the size of each list. We claim that $|L_i| = O(n \log(nT)/\varepsilon)$ for all i. The values in L_i are sorted, the smallest is 0, and consecutive values differ by a factor of at least $(1+\delta)$. If $L_i = \{s_1, s_2, \ldots, s_k\}$ with $0 = s_1 < s_2 < \cdots < s_k$, then $s_2 \ge 1$ and $s_j \ge (1+\delta)^{j-2}$ for $j \ge 2$. Since $s_k \le \sum_{i=1}^{n} a_i \le nT$, we have $(1+\delta)^{k-2} \le nT$ and thus, taking logarithms, $(k-2)\log(1+\delta) \le \log(nT)$. Using $\log(1+\delta) \ge \delta/2$ for $\delta \le 1$, we get:

$$|L_i| = k \le 2 + \frac{\log(nT)}{\delta/2} = O\left(\frac{n \log(nT)}{\varepsilon}\right)$$

For the running time, each iteration i processes the list L_{i-1}, whose size is $O(n \log(nT)/\varepsilon)$. Computing L'_i takes $O(|L_{i-1}|)$ time, trimming takes $O(|L'_i|)$ time, and sorting takes $O(|L'_i| \log |L'_i|) =$

$O((n\log(nT)/\varepsilon)\log(nT))$ time. Over n iterations, the total time is:

$$O\left(n \cdot \frac{n\log(nT)}{\varepsilon}\log(nT)\right) = O\left(\frac{n^2\log^2(nT)}{\varepsilon}\right)$$

This concludes the proof. □

September 7, 2026

Lecture 2: Approximation Algorithms II

In the previous lecture, we studied approximation algorithms for vertex cover, set cover, and subset sum. We saw constant-factor approximations, logarithmic approximations, and an FPTAS. In this lecture, we continue with three more problems. The metric traveling salesman problem illustrates how structural assumptions (the triangle inequality) can turn an inapproximable problem into one admitting a constant-factor guarantee. The knapsack problem gives a second example of an FPTAS, this time via profit scaling rather than trimming. Load balancing on identical machines shows how a simple algorithmic refinement (sorting the input) can dramatically improve an approximation ratio.

2.1 Metric Traveling Salesman Problem

The traveling salesman problem (TSP) asks for a minimum-cost tour that visits every vertex in a graph exactly once and returns to the start. Given a complete graph K_n on n vertices with edge costs $c(u, v) \geq 0$, we seek a Hamiltonian cycle of minimum total cost. In the general case, TSP is extremely hard to approximate. Sahni and Gonzalez [7] showed that no finite approximation ratio is achievable.

Theorem 2.1. Unless P = NP, for any polynomial-time computable function $\rho(n)$, there is no polynomial-time $\rho(n)$-approximation for the general TSP.

Proof. We reduce from the Hamiltonian Cycle problem. Given a graph $G = (V, E)$ on n vertices, construct a complete graph K_n on the same vertex set with costs:

$$c(u, v) = \begin{cases} 1 & \text{if } \{u, v\} \in E, \\ n \cdot \rho(n) + 1 & \text{otherwise.} \end{cases}$$

If G has a Hamiltonian cycle, the optimal TSP tour has cost n. If G has no Hamiltonian cycle, any tour must use at least one non-edge, so the optimal tour has cost at least $n - 1 + n\rho(n) + 1 = n \cdot \rho(n) + n > \rho(n) \cdot n$.

A $\rho(n)$-approximation algorithm would produce a tour of cost at most $\rho(n) \cdot n$ when OPT $= n$, distinguishing the two cases. Since Hamiltonian Cycle is NP-complete, no such algorithm exists unless P = NP. □

Why is the general case so hopeless? The reduction above constructs instances where non-edges receive artificially enormous costs, creating a gap between "yes" and "no" instances that no algorithm can bridge. The triangle inequality prevents exactly this kind of construction: if there is a path of cost k between u and w, then the direct edge $c(u, w)$ can cost at most k. Formally, the metric TSP requires edge costs to satisfy:

$$c(u, w) \leq c(u, v) + c(v, w) \quad \text{for all } u, v, w \in V.$$

This restriction is natural in many applications (e.g., Euclidean distances). It does not, however, make the problem easy: metric TSP is still NP-hard. The same reduction from Hamiltonian Cycle works, only with a bounded cost gap: put $c(u, v) = 1$ if $\{u, v\} \in E$ and $c(u, v) = 2$ otherwise. Every cost now lies in $\{1, 2\}$, so $c(u, w) \leq 2 \leq c(u, v) + c(v, w)$ holds for all $u, v, w \in V$ and the instance is metric. If G has a Hamiltonian cycle, the optimal tour costs exactly n; otherwise every tour uses a non-edge and costs at least $n + 1$. An exact algorithm for metric TSP would thus decide Hamiltonian Cycle.

For approximation, the triangle inequality has a crucial algorithmic consequence: *shortcuts are free*. If a walk visits vertices $u, v_1, \ldots, v_k, w$, we can skip the intermediate vertices and go directly from u to w

without increasing the cost, since $c(u, w) \leq c(u, v_1) + c(v_1, v_2) + \cdots + c(v_k, w)$. This means that any closed walk through all vertices can be converted into a Hamiltonian cycle of no greater cost. So the problem reduces to: find a cheap closed walk that visits every vertex. This is easier than finding a Hamiltonian cycle directly, because walks are allowed to revisit vertices.

As in the vertex cover analysis from the previous lecture, we start by looking for a good lower bound on OPT. An optimal tour visits all n vertices and returns to the start. Removing one edge from this tour yields a spanning path, which is in particular a spanning tree. So every tour contains a spanning tree, and the minimum spanning tree gives a lower bound. Crucially, MSTs can be computed efficiently: sort the edges by cost and greedily add each edge that does not create a cycle (Kruskal's algorithm), which runs in $O(m \log n)$ time.

Lemma 2.1. Let T be a minimum spanning tree of K_n. Then $c(T) \leq \text{OPT}$.

Proof. Removing any edge from an optimal tour yields a spanning tree of cost at most OPT. Since T is a minimum spanning tree, $c(T)$ is at most the cost of any spanning tree, so $c(T) \leq \text{OPT}$. □

Now the strategy is clear: use the MST as a backbone and turn it into a closed walk (which we can then shortcut into a tour). A tree is not a closed walk, but we can make it one by traversing each edge twice, once in each direction. This gives a multigraph where every vertex has even degree, which admits an Eulerian tour. Such a tour can be found in linear time: start at any vertex and greedily follow unused edges until returning to the start; whenever the resulting circuit misses some edges, splice in a sub-circuit from a vertex that still has unused edges, and repeat until all edges are used.

Algorithm (MST-doubling):

1. Compute a minimum spanning tree T of K_n.
2. Double every edge of T to obtain a multigraph H where every vertex has even degree.
3. Find an Eulerian tour of H.
4. Shortcut the Eulerian tour: skip any vertex already visited, going directly to the next unvisited vertex.

Theorem 2.2. The MST-doubling algorithm is a 2-approximation for metric TSP.

Proof. The Eulerian tour of H has cost $c(H) = 2 \cdot c(T) \leq 2 \cdot \text{OPT}$ by Lemma 2.1. Shortcutting can only decrease the cost by the triangle inequality. Therefore, the final tour has cost at most $2 \cdot \text{OPT}$. □

The factor of 2 comes from doubling *every* edge of the MST. But why did we double edges in the first place? Only to make all degrees even, so that an Eulerian tour exists. If a vertex already has even degree, doubling its incident edges is wasteful. The vertices that actually need fixing are precisely those with *odd* degree. The key idea, due to Christofides [6] and independently Serdyukov [30], is to add edges only where needed: a minimum-weight perfect matching on the odd-degree vertices.

Algorithm (Christofides–Serdyukov):

1. Compute a minimum spanning tree T of K_n.
2. Let S be the set of vertices with odd degree in T.
3. Compute a minimum-weight perfect matching M on the complete graph induced by S.

4. Combine T and M to form a multigraph $H = T \cup M$.
5. Find an Eulerian tour of H and shortcut repeated vertices.

Why does a perfect matching on S exist? The sum of degrees in any graph is even, so the number of odd-degree vertices $|S|$ is even. Since we compute the matching on the *complete* graph induced by S, any pairing of the $|S|$ vertices into $|S|/2$ pairs gives a perfect matching. Finding a minimum-weight such matching is a nontrivial algorithmic problem: the main difficulty is that, unlike in bipartite graphs, augmenting paths in general graphs can traverse odd cycles, which Edmonds [33] resolved by "shrinking" these cycles (called *blossoms*) into single vertices. The resulting algorithm runs in $O(n^3)$ time. Adding the matching edges to T increases the degree of each vertex in S by exactly one, making all degrees even and enabling the Eulerian tour.

Lemma 2.2. The minimum-weight perfect matching M on S satisfies $c(M) \le \mathrm{OPT}/2$.

Proof. Consider an optimal TSP tour and restrict it to the vertices in S. Since the tour visits every vertex, we can shortcut it to obtain a tour on S alone. By the triangle inequality, this restricted tour has cost at most OPT.

Let $|S| = 2k$. The restricted tour on S visits $2k$ vertices in some order $s_1, s_2, \ldots, s_{2k}, s_1$. Partition the edges of this tour into two perfect matchings:

$$M_1 = \{\{s_1, s_2\}, \{s_3, s_4\}, \ldots, \{s_{2k-1}, s_{2k}\}\}$$

$$M_2 = \{\{s_2, s_3\}, \{s_4, s_5\}, \ldots, \{s_{2k}, s_1\}\}$$

Both M_1 and M_2 are perfect matchings on S, and their total cost satisfies $c(M_1) + c(M_2) \le \mathrm{OPT}$. Therefore, $\min(c(M_1), c(M_2)) \le \mathrm{OPT}/2$. Since M is a minimum-weight perfect matching, $c(M) \le \mathrm{OPT}/2$. □

Theorem 2.3. The Christofides–Serdyukov algorithm is a $\frac{3}{2}$-approximation for metric TSP.

Proof. The multigraph $H = T \cup M$ has cost $c(H) = c(T) + c(M)$. By Lemma 2.1, $c(T) \le \mathrm{OPT}$. By Lemma 2.2, $c(M) \le \mathrm{OPT}/2$. Therefore:

$$c(H) = c(T) + c(M) \le \mathrm{OPT} + \frac{\mathrm{OPT}}{2} = \frac{3}{2} \cdot \mathrm{OPT}$$

As in the MST-doubling analysis, shortcutting the Eulerian tour of H can only decrease the cost by the triangle inequality. The final tour has cost at most $\frac{3}{2} \cdot \mathrm{OPT}$. □

The $\frac{3}{2}$-approximation ratio stood as the best known for nearly fifty years. In 2020, Karlin, Klein, and Oveis Gharan [31] achieved a breakthrough by obtaining a $(\frac{3}{2} - \varepsilon_0)$-approximation for some small constant $\varepsilon_0 > 0$. On the hardness side, metric TSP is known to be APX-hard: there is no PTAS unless P = NP.

2.2 Knapsack

The knapsack problem is one of the most fundamental problems in combinatorial optimization. We are given n items, where item i has weight $w_i > 0$ and profit $p_i > 0$, along with a knapsack capacity W. The goal is to find a subset $S \subseteq \{1, \ldots, n\}$ maximizing $\sum_{i \in S} p_i$ subject to $\sum_{i \in S} w_i \le W$. We may assume

without loss of generality that $w_i \le W$ for all i, since items that do not fit individually can be discarded.

Subset sum is the special case where $w_i = p_i$ for all i and W is the target value. In the previous lecture, we gave an FPTAS for subset sum using a trimming technique. Here we present a different approach to an FPTAS for the more general knapsack problem, based on profit scaling. The idea is again quite simple: we already know how to solve knapsack *exactly* by dynamic programming, but the running time depends on the size of the profit values. If we could make these values smaller, the DP would be faster. Rounding profits down loses some precision, but if we control the rounding error, the solution remains close to optimal.

The standard DP for knapsack indexes by weight and runs in time $O(nW)$. For our purposes, we index by *profit* instead. Let $A[i][p]$ be the minimum weight to achieve profit exactly p using items from $\{1, \dots, i\}$. The recurrence is:

$$A[i][p] = \min\big(A[i-1][p],\ A[i-1][p-p_i] + w_i\big)$$

with base cases $A[0][0] = 0$ and $A[0][p] = \infty$ for $p > 0$. The optimal profit is the largest p such that $A[n][p] \le W$. Let $P = \sum_{i=1}^{n} p_i$ denote the total profit. The table has $O(nP)$ entries, so this DP runs in $O(nP)$ time, which is pseudo-polynomial: polynomial in the *values* of the profits but potentially exponential in the input size. Why index by profit rather than weight? Because we are about to *scale down* the profits, making P polynomial in n and $1/\varepsilon$. We have no such control over W. How much can we afford to round? Replacing each profit p_i by $\lfloor p_i/K \rfloor$ for some scaling factor K introduces a per-item error of at most K. Over at most n items, the total error is at most nK. For this to be at most $\varepsilon \cdot \mathrm{OPT}$, we need $nK \le \varepsilon \cdot \mathrm{OPT}$. We don't know OPT, but we do know that $\mathrm{OPT} \ge p_{\max}$ (since every item fits individually). Setting $K = \varepsilon \cdot p_{\max}/n$ guarantees $nK = \varepsilon \cdot p_{\max} \le \varepsilon \cdot \mathrm{OPT}$.

Algorithm (FPTAS for Knapsack):

1. Let $p_{\max} \leftarrow \max_i p_i$.
2. Set the scaling factor $K \leftarrow \frac{\varepsilon \cdot p_{\max}}{n}$.
3. For each item i, define the scaled profit $p'_i \leftarrow \left\lfloor \frac{p_i}{K} \right\rfloor$.
4. Run the exact DP on the scaled profits $p'_1, \dots, p'_n$ with the original weights $w_1, \dots, w_n$ and capacity W.
5. Return the subset found by the DP.

Theorem 2.4. For any $\varepsilon > 0$, the above algorithm is a $(1+\varepsilon)$-approximation for knapsack, running in time $O(n^3/\varepsilon)$.

Proof. Each scaled profit satisfies:

$$p'_i = \left\lfloor \frac{p_i}{K} \right\rfloor \le \frac{p_i}{K} \le \frac{p_{\max}}{K} = \frac{n}{\varepsilon}$$

The total scaled profit is $P' = \sum_{i=1}^{n} p'_i \le n^2/\varepsilon$. The DP runs in $O(nP') = O(n^3/\varepsilon)$ time, which is polynomial in n and $1/\varepsilon$.

For the approximation guarantee, let S^* be an optimal solution with profit $\mathrm{OPT} = \sum_{i \in S^*} p_i$. Since S^* satisfies the weight constraint, the DP on scaled profits finds a solution S with $\sum_{i \in S} p'_i \ge \sum_{i \in S^*} p'_i$.

For each item i, the floor satisfies $p'_i \ge p_i/K - 1$, so $p'_i \cdot K \ge p_i - K$. Therefore the true profit of S satisfies:

$$\sum_{i \in S} p_i \ge \sum_{i \in S} p'_i \cdot K \ge \sum_{i \in S^*} p'_i \cdot K \ge \sum_{i \in S^*} (p_i - K) = \mathrm{OPT} - |S^*| \cdot K$$

Since $|S^*| \le n$ and $K = \varepsilon \cdot p_{\max}/n$, the total rounding error is $|S^*| \cdot K \le \varepsilon \cdot p_{\max} \le \varepsilon \cdot \mathrm{OPT}$. Hence $\mathrm{ALG} \ge (1-\varepsilon) \cdot \mathrm{OPT}$. To obtain a $(1+\varepsilon)$-approximation, we run the algorithm with parameter $\varepsilon/2$ instead of ε. This gives $\mathrm{ALG} \ge (1-\varepsilon/2) \cdot \mathrm{OPT} \ge \mathrm{OPT}/(1+\varepsilon)$, since $(1-\varepsilon/2)(1+\varepsilon) = 1+\varepsilon/2-\varepsilon^2/2 \ge 1$ for $\varepsilon \le 1$. The running time becomes $O(n^3/(\varepsilon/2)) = O(n^3/\varepsilon)$. □

The profit-scaling technique, introduced by Ibarra and Kim [8], is conceptually simpler than the trimming approach we saw for subset sum. Both achieve an FPTAS, but scaling has the advantage of reducing to an existing exact algorithm as a black box: we simply change the input and run the same DP.

2.3 Load Balancing

We now turn to a scheduling problem. We have m identical machines and n jobs with processing times $t_1, t_2, \ldots, t_n$. Each job must be assigned to exactly one machine, and the load of a machine is the total processing time of jobs assigned to it. The objective is to minimize the *makespan*: the maximum load over all machines.

Formally, given an assignment $\sigma \colon \{1, \ldots, n\} \to \{1, \ldots, m\}$, the makespan is:

$$\max_{1 \le j \le m} \sum_{i : \sigma(i) = j} t_i$$

As before, we start by identifying lower bounds on the optimal makespan. Two are immediate.

Lemma 2.3. The optimal makespan satisfies:

$$\mathrm{OPT} \ge \frac{1}{m} \sum_{i=1}^{n} t_i \quad \text{and} \quad \mathrm{OPT} \ge \max_{1 \le i \le n} t_i$$

Proof. The total work $\sum_i t_i$ is distributed among m machines, so by averaging the most loaded machine has load at least $\frac{1}{m} \sum_i t_i$. The second bound holds because whichever machine processes the longest job has load at least $\max_i t_i$. □

The simplest scheduling heuristic is list scheduling, introduced by Graham [9]: process jobs one by one, always assigning the next job to the least loaded machine. This is the greedy algorithm that tries to keep loads balanced at every step.

Algorithm (List Scheduling):

1. Process jobs in the order $1, 2, \ldots, n$.
2. Assign each job to the machine with the current smallest load.

To analyze list scheduling, we focus on the bottleneck machine and the moment the last job lands on it. The makespan decomposes as the load L before that job plus the job's processing time t_ℓ. We can bound each term separately using our two lower bounds.

Theorem 2.5. List scheduling is a $(2 - 1/m)$-approximation for makespan minimization.

Proof. Let M_j be the machine with the maximum load at the end, let t_ℓ be the last job assigned to it, and let L be the load of M_j just before job ℓ was assigned. The makespan is $L + t_\ell$.

When job ℓ was assigned, M_j had the smallest load among all machines, so every machine had load at least L. The total load on all m machines at that point is $\sum_{i<\ell} t_i \le \sum_i t_i - t_\ell$, and since M_j has the smallest load, $m \cdot L \le \sum_i t_i - t_\ell$. Therefore:

$$L \le \frac{1}{m}\left(\sum_{i=1}^{n} t_i - t_\ell\right) \le \mathrm{OPT} - \frac{t_\ell}{m}$$

by Lemma 2.3. Also, $t_\ell \le \max_i t_i \le \mathrm{OPT}$ by the same lemma. Therefore:

$$\begin{aligned}\mathrm{ALG} = L + t_\ell &\le \mathrm{OPT} - \frac{t_\ell}{m} + t_\ell = \mathrm{OPT} + \left(1 - \frac{1}{m}\right) t_\ell \\ &\le \mathrm{OPT} + \left(1 - \frac{1}{m}\right)\mathrm{OPT} = \left(2 - \frac{1}{m}\right)\mathrm{OPT}\end{aligned}$$

This concludes the proof. □

The bound $(2 - 1/m)$ is tight for list scheduling. The worst case arises when a large job arrives last and gets piled onto an already loaded machine. Consider $m(m-1)$ jobs of size 1 followed by one job of size m. List scheduling distributes the small jobs evenly, giving each machine load $m - 1$, then assigns the large job to one of them, resulting in makespan $2m - 1$. The optimal schedule places the large job alone on one machine and distributes the small jobs among the remaining $m - 1$ machines, achieving makespan m. The ratio is $(2m-1)/m = 2 - 1/m$.

The problem is clear: list scheduling is oblivious to job sizes, so a large job can arrive last and ruin the balance. The fix is equally clear: process large jobs first, when the machines are still lightly loaded. Graham [10] observed that this simple modification leads to a significantly improved guarantee.

Algorithm (Longest Processing Time first, LPT):

1. Sort jobs so that $t_1 \ge t_2 \ge \cdots \ge t_n$.
2. Apply list scheduling in this order.

The key consequence of sorting is that the last job assigned to the bottleneck machine is now the *smallest* job involved, not an arbitrarily large one. This lets us replace the bound $t_\ell \le \mathrm{OPT}$ with the much tighter $t_\ell \le \mathrm{OPT}/3$.

Theorem 2.6. LPT is a $(4/3 - 1/(3m))$-approximation for makespan minimization.

Proof. If $n \le m$, each job gets its own machine and the makespan equals $\max_i t_i = \mathrm{OPT}$. So assume $n > m$.

Let M_j be the bottleneck machine and let t_ℓ be the last job assigned to it. If M_j received only one job, then $\mathrm{ALG} = t_\ell \le \mathrm{OPT}$ and we are done. Otherwise, M_j received at least two jobs, so $\ell \ge m + 1$ (the first m jobs are assigned to distinct machines), and therefore $t_\ell \le t_{m+1}$.

We distinguish two cases.

Case 1: $t_\ell \le \mathrm{OPT}/3$. Using the same bound on L as in Theorem 2.5:

$$\begin{aligned}\mathrm{ALG} = L + t_\ell &\le \mathrm{OPT} - \frac{t_\ell}{m} + t_\ell = \mathrm{OPT} + \left(1 - \frac{1}{m}\right) t_\ell \\ &\le \mathrm{OPT} + \left(1 - \frac{1}{m}\right) \frac{\mathrm{OPT}}{3} = \left(\frac{4}{3} - \frac{1}{3m}\right) \mathrm{OPT}\end{aligned}$$

Case 2: $t_\ell > \mathrm{OPT}/3$. Since the jobs are sorted, $t_1 \ge \cdots \ge t_\ell > \mathrm{OPT}/3$, so no machine of an optimal schedule holds three of the jobs $1, \ldots, \ell$; in particular $\ell \le 2m$. Consider the moment LPT assigns job ℓ to M_j (Figure 2.1, left). The $\ell - 1 \ge m$ jobs assigned so far occupy every machine, since LPT fills empty machines first. Let k be the number of machines holding exactly one of them. The other $m - k$ machines hold at least two, so $\ell - 1 \ge k + 2(m-k)$, that is, $k \ge 2m - \ell + 1 \ge 1$. Since M_j has the smallest load, each of these k singleton jobs has processing time at least L.

Now consider an optimal schedule (Figure 2.1, right). If each of the k singleton jobs were the only job among $1, \ldots, \ell$ on its machine, the remaining $\ell - k$ of these jobs would lie on the remaining $m - k$ machines, at most two per machine, giving $\ell - k \le 2(m-k)$, that is, $k \le 2m - \ell$. This contradicts $k \ge 2m - \ell + 1$. Hence some machine of the optimal schedule holds a singleton job, of processing time at least L, together with another job of index at most ℓ, of processing time at least t_ℓ. Therefore $\mathrm{OPT} \ge L + t_\ell = \mathrm{ALG}$.

In both cases, $\mathrm{ALG} \le (4/3 - 1/(3m)) \cdot \mathrm{OPT}$. □

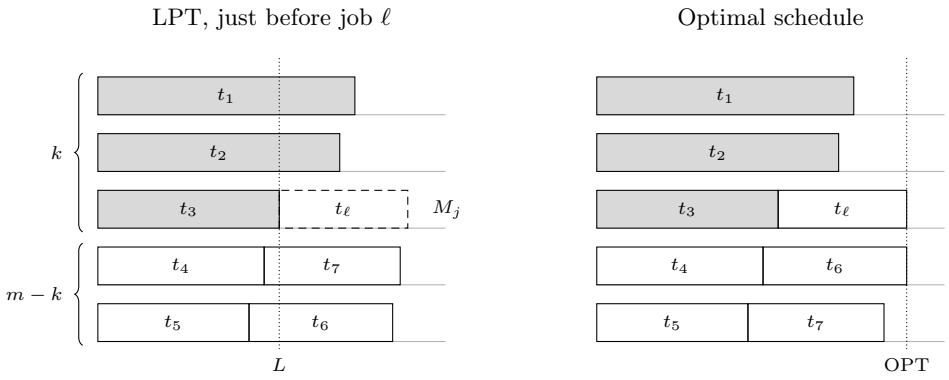


Figure 2.1: Case 2 in the proof of Theorem 2.6, with $m = 5$ and $\ell = 8$.

The 4/3 ratio of LPT is a significant improvement over the factor of 2 from plain list scheduling, obtained simply by sorting the input. For the load balancing problem, Hochbaum and Shmoys [32] showed that a PTAS exists: for any $\varepsilon > 0$, a $(1+\varepsilon)$-approximation can be found in polynomial time, though the algorithm is considerably more involved than the simple LPT rule.

September 14, 2026

Lecture 3: Approximation Algorithms III

In the previous two lectures, we designed approximation algorithms using combinatorial techniques: greedy algorithms, matchings, dynamic programming, and scaling. In this lecture, we add two more tools to our repertoire. First, we introduce linear programming (LP) relaxation: formulate the problem as an integer program, relax the integrality constraints, solve the resulting LP, and round the fractional solution. We illustrate this on weighted vertex cover and weighted set cover. Then, we study MAX-CUT via semidefinite programming (SDP), a powerful generalization of LP relaxation.

3.1 Linear Programming Relaxation

A linear program (LP) asks to optimize a linear objective function subject to linear constraints. In its standard form, an LP minimizes or maximizes ${}^t c\, x$ subject to $Ax \leq b$ and $x \geq 0$, where $x \in \mathbb{R}^n$. Linear programs can be solved in polynomial time, for example by interior point methods.

An integer linear program (ILP) is an LP with the additional constraint that the variables must take integer values: $x \in \mathbb{Z}^n$. In general, solving an ILP is NP-hard. For combinatorial optimization problems, the variables are usually binary ($x \in \{0,1\}^n$), and we focus on this case.

The key idea behind LP relaxation is to drop the integrality constraints, obtaining a problem that is easier to solve and whose optimal value provides a bound on the true optimum.

Definition 3.1: LP relaxation

Given an ILP with binary variables $x_i \in \{0,1\}$, its LP relaxation is obtained by replacing each integrality constraint with the continuous constraint $0 \leq x_i \leq 1$.

For a minimization problem, the LP relaxation has a larger feasible region (every integer solution is also a fractional solution), so $\mathrm{OPT}_{\mathrm{LP}} \leq \mathrm{OPT}$. For a maximization problem, $\mathrm{OPT}_{\mathrm{LP}} \geq \mathrm{OPT}$. In both cases, the LP optimum provides a bound on the true optimum that we can exploit algorithmically.

Definition 3.2: Integrality gap

The integrality gap of an LP relaxation is the worst-case ratio between the optimal integer solution and the optimal fractional solution, taken over all instances. For a minimization problem, it is $\sup_I \frac{\mathrm{OPT}(I)}{\mathrm{OPT}_{\mathrm{LP}}(I)}$; for a maximization problem, it is $\inf_I \frac{\mathrm{OPT}(I)}{\mathrm{OPT}_{\mathrm{LP}}(I)}$.

The integrality gap characterizes the power of a relaxation: no rounding scheme can achieve an approximation ratio better than the integrality gap, and conversely, if a rounding scheme achieves cost at most $\alpha \cdot \mathrm{OPT}_{\mathrm{LP}}$ on every instance (for a minimization problem), then it is an α-approximation algorithm, since $\mathrm{OPT}_{\mathrm{LP}} \leq \mathrm{OPT}$. The integrality gap is therefore exactly the best approximation ratio achievable by rounding a given relaxation.

Not all LP relaxations are useful. For example, the natural LP relaxation for maximum-weight independent set (maximize $\sum_v w_v x_v$ subject to $x_u + x_v \leq 1$ for all edges, $0 \leq x_v \leq 1$) has an integrality gap of $n/2$ on the complete graph K_n: the LP sets every $x_v = 1/2$ for a value of $n/2$, while the integer optimum is 1. No rounding scheme can overcome such a gap. The problems below have much better integrality gaps.

3.2 Weighted Vertex Cover

We now revisit the vertex cover problem, this time in its weighted version. Given an undirected graph $G = (V, E)$ and vertex weights $w_v \geq 0$ for each $v \in V$, the goal is to find a vertex cover $C \subseteq V$ (a set such that

every edge has at least one endpoint in C) of minimum total weight $w(C) = \sum_{v \in C} w_v$.

In Lecture 1, the edge-picking algorithm (Theorem 1.3) gave a 2-approximation for unweighted vertex cover. However, that combinatorial approach does not directly extend to the weighted case, since picking both endpoints of an edge may include a very heavy vertex unnecessarily. LP relaxation provides a clean way to handle weights.

We formulate the problem as an ILP. For each vertex $v \in V$, introduce a variable $x_v \in \{0, 1\}$ indicating whether v is in the cover. The ILP is:

$$\begin{aligned} \text{minimize} \quad & \sum_{v \in V} w_v \, x_v \\ \text{subject to} \quad & x_u + x_v \geq 1 && \text{for all } \{u, v\} \in E \\ & x_v \in \{0, 1\} && \text{for all } v \in V \end{aligned}$$

The LP relaxation replaces $x_v \in \{0, 1\}$ with $0 \leq x_v \leq 1$:

$$\begin{aligned} \text{minimize} \quad & \sum_{v \in V} w_v \, x_v \\ \text{subject to} \quad & x_u + x_v \geq 1 && \text{for all } \{u, v\} \in E \\ & 0 \leq x_v \leq 1 && \text{for all } v \in V \end{aligned} \tag{1}$$

Let x^* be an optimal solution to the LP relaxation. We round this fractional solution to an integer solution by a simple threshold rule.

Algorithm:

1. Solve the LP relaxation (1) to obtain an optimal solution x^*.
2. For each vertex $v \in V$: include v in C if and only if $x_v^* \geq 1/2$.
3. Return C.

Theorem 3.1. The LP rounding algorithm is a 2-approximation for weighted vertex cover.

Proof. We must verify two things: that C is a valid vertex cover, and that its weight is at most $2 \cdot \text{OPT}$.

For feasibility, consider any edge $\{u, v\} \in E$. The LP constraint gives $x_u^* + x_v^* \geq 1$. Therefore, at least one of x_u^* or x_v^* is at least $1/2$, so at least one of u or v is included in C. Hence C is a vertex cover.

For the cost bound, for every vertex $v \in C$, we have $x_v^* \geq 1/2$, so $1 \leq 2x_v^*$. Therefore:

$$w(C) = \sum_{v \in C} w_v \leq \sum_{v \in C} w_v \cdot 2x_v^* \leq 2 \sum_{v \in V} w_v \, x_v^* = 2 \cdot \text{OPT}_{\text{LP}} \leq 2 \cdot \text{OPT}$$

This completes the proof. □

We now show that the factor of 2 is tight for this LP relaxation, meaning no rounding scheme can do better.

Theorem 3.2. The integrality gap of the vertex cover LP relaxation (1) is exactly 2.

Proof. Theorem 3.1 shows that the integrality gap is at most 2. For the lower bound, consider the complete graph K_n with unit weights $w_v = 1$ for all v. The fractional solution $x_v^* = 1/2$ for all v is feasible: for every edge $\{u, v\}$, we have $x_u^* + x_v^* = 1 \geq 1$. Its cost is $\mathrm{OPT}_{\mathrm{LP}} \leq n/2$. On the other hand, any integer vertex cover must include at least $n - 1$ vertices (omitting at most one vertex), so $\mathrm{OPT} = n - 1$. The ratio $\mathrm{OPT}/\mathrm{OPT}_{\mathrm{LP}} \geq (n-1)/(n/2) = 2(n-1)/n$, which approaches 2 as $n \to \infty$. □

We remark that the vertex cover LP has a remarkable structural property: every basic feasible solution of the LP relaxation (1) is half-integral, meaning that every variable takes a value in $\{0, 1/2, 1\}$. This is a consequence of the total dual integrality of the constraint matrix and explains why the simple threshold $1/2$ works so well.

3.3 Weighted Set Cover

In the weighted set cover problem, we are given a universe $U = \{e_1, \ldots, e_n\}$ of n elements, a collection $\mathcal{S} = \{S_1, \ldots, S_m\}$ of subsets of U, and a cost $c_j \geq 0$ for each set S_j. The goal is to find a subcollection of minimum total cost that covers every element, i.e., whose union is U. In Lecture 1, we showed that the greedy algorithm is an H_n-approximation (Theorem 1.4), and that this is essentially optimal [3]. LP relaxation gives a different, incomparable guarantee.

The *frequency* of an element e is the number of sets containing it: $f_e = |\{j : e \in S_j\}|$. Let $f = \max_e f_e$ be the maximum frequency.

We formulate the problem as an ILP with a variable $x_j \in \{0, 1\}$ for each set S_j:

$$\begin{aligned} \text{minimize} \quad & \sum_{j=1}^{m} c_j \, x_j & & \\ \text{subject to} \quad & \sum_{j : e \in S_j} x_j \geq 1 & & \text{for all } e \in U \\ & x_j \in \{0, 1\} & & \text{for all } j \end{aligned} \tag{2}$$

The LP relaxation replaces $x_j \in \{0, 1\}$ with $0 \leq x_j \leq 1$. Let x^* be an optimal LP solution with cost $\mathrm{OPT}_{\mathrm{LP}} \leq \mathrm{OPT}$.

Algorithm:

1. Solve the LP relaxation to obtain x^*.
2. Include S_j in the cover if and only if $x_j^* \geq 1/f$.
3. Return the selected sets.

Theorem 3.3. The LP rounding algorithm is an f-approximation for set cover.

Proof. For feasibility, consider any element $e \in U$. The LP constraint gives $\sum_{j:e\in S_j} x_j^* \geq 1$. This sum has at most $f_e \leq f$ terms, so at least one of them satisfies $x_j^* \geq 1/f$. That set is included, and e is covered.

For the cost bound, every selected set has $x_j^* \geq 1/f$, so $1 \leq f \cdot x_j^*$. Therefore:

$$\sum_{j\in\text{cover}} c_j \leq \sum_{j\in\text{cover}} c_j \cdot f \cdot x_j^* \leq f \sum_{j=1}^{m} c_j\, x_j^* = f \cdot \text{OPT}_{\text{LP}} \leq f \cdot \text{OPT}$$

This concludes the proof. □

When f is a small constant, this gives a constant-factor approximation. Vertex cover is the special case where $U = E$, each set $S_v = \{e \in E : v \in e\}$ corresponds to a vertex, and every element has frequency exactly 2; the f-approximation recovers the factor of 2. When f is large, the greedy H_n-approximation from Lecture 1 is better.

We proved the greedy H_n-approximation combinatorially in Lecture 1. LP duality allows us to reinterpret that proof and determine the integrality gap of the set cover LP. Every LP has an associated *dual* program. For a primal LP in the form

$$\text{minimize } {}^t c\, x \text{ subject to } Ax \geq b,\ x \geq 0,$$

the dual is

$$\text{maximize } {}^t b\, y \text{ subject to } {}^t A\, y \leq c,\ y \geq 0.$$

The dual has one variable per primal constraint and one constraint per primal variable.

Theorem 3.4 (Weak duality). If x is primal feasible and y is dual feasible, then ${}^t c\, x \geq {}^t b\, y$.

Proof. Since ${}^t A\, y \leq c$ and $x \geq 0$, we have ${}^t c\, x \geq {}^t({}^t A\, y)\, x = {}^t y\, Ax$. Since $Ax \geq b$ and $y \geq 0$, we have ${}^t y\, Ax \geq {}^t y\, b = {}^t b\, y$. □

In other words, every dual feasible solution provides a lower bound on the primal optimum (for minimization problems). We do not need to solve the LP to use this: any feasible dual solution, however obtained, gives a certifiable lower bound.

The dual of the set cover LP (2) introduces a variable $y_e \geq 0$ for each element $e \in U$:

$$\begin{aligned} \text{maximize} \quad & \sum_{e\in U} y_e & & \\ \text{subject to} \quad & \sum_{e\in S_j} y_e \leq c_j & & \text{for all } j \\ & y_e \geq 0 & & \text{for all } e \in U \end{aligned}$$

This is a *packing* problem: assign nonnegative values to elements so that no set is overloaded, and maximize the total value. By weak duality, any feasible packing gives a lower bound: $\sum_e y_e \leq \text{OPT}$.

The greedy algorithm from Lecture 1 implicitly constructs a near-feasible dual solution. Recall that the greedy analysis assigns a price $\text{price}(e)$ to each element e when it is first covered: if e is covered by a set S_j that covers k new elements at that step, then $\text{price}(e) = c_j/k$. Setting $y_e = \text{price}(e)$, the total cost of the greedy solution equals $\sum_{e\in U} y_e$. This y is not dual feasible in general: for a set S_j, we may have

$\sum_{e \in S_j} y_e > c_j$. However, the greedy proof (Theorem 1.4) shows that $\sum_{e \in S_j} y_e \leq H_n \cdot c_j$ for every set S_j. Therefore, y/H_n is dual feasible, giving:

$$\frac{1}{H_n} \sum_{e \in U} y_e = \sum_{e \in U} \frac{y_e}{H_n} \leq \text{OPT}$$

Hence $\text{ALG} = \sum_e y_e \leq H_n \cdot \text{OPT}_{\text{LP}} \leq H_n \cdot \text{OPT}$. The worst-case approximation ratio is the same as in Lecture 1, but we learn something new: the integrality gap of the set cover LP is at least H_n. Indeed, the dual solution has value ALG, so $\text{OPT}_{\text{LP}} \geq \text{ALG}/H_n$ by weak duality, and there are instances where $\text{ALG} = H_n \cdot \text{OPT}$. Combined with the f-approximation from LP rounding (Theorem 3.3), this shows that the LP relaxation captures the approximability of set cover: the gap between the LP and integer optima is $\Theta(H_n)$ in the worst case, matching the best possible approximation ratio. This technique of building an approximately feasible dual solution and rescaling it is called *dual fitting*.

3.4 MAX-CUT

We now turn to a maximization problem. Given an undirected graph $G = (V, E)$ with nonnegative edge weights $w_{ij} \geq 0$, the MAX-CUT problem asks for a partition of V into two sets S and $\bar{S} = V \setminus S$ maximizing the total weight of edges crossing the cut: $w(S) = \sum_{\{i,j\} \in E,\, i \in S,\, j \notin S} w_{ij}$. We write $W = \sum_{\{i,j\} \in E} w_{ij}$ for the total edge weight; clearly $w(S) \leq W$ for every cut. MAX-CUT is NP-hard. Moreover, it is APX-hard: no PTAS exists unless P = NP.

A simple algorithm assigns each vertex to S independently with probability $1/2$.

Theorem 3.5. The randomized algorithm satisfies $\mathbb{E}[w(S)] \geq \frac{1}{2} W \geq \frac{1}{2} \cdot \text{OPT}$.

Proof. By linearity of expectation:

$$\mathbb{E}[w(S)] = \sum_{\{i,j\} \in E} w_{ij} \cdot \mathbb{P}[i \text{ and } j \text{ on different sides}] = \sum_{\{i,j\} \in E} w_{ij} \cdot \frac{1}{2} = \frac{W}{2}.$$

Since $\text{OPT} \leq W$, we have $\mathbb{E}[w(S)] \geq \frac{1}{2} \cdot \text{OPT}$. □

This can be derandomized by the *method of conditional expectations*. The idea is to fix the random choices one at a time, each time choosing the value that does not decrease the conditional expectation of the cut weight. The expectation is $W/2$ before anything is fixed, and once everything is fixed, the expectation is just the weight of the resulting cut, which is therefore at least $W/2$. The method works because these conditional expectations can be computed exactly.

Order the vertices as $1, \ldots, n$ and let $Z_i \in \{0, 1\}$ indicate whether vertex i is placed in S. Suppose the first k vertices have been assigned values $z_1, \ldots, z_k$. An edge with both endpoints among these vertices is cut or not, depending on whether $z_i \neq z_j$. An edge with at least one endpoint not yet assigned is cut with probability exactly $1/2$, since that endpoint is placed uniformly at random, independently of everything else. Hence

$$\mathbb{E}[w(S) \mid Z_1 = z_1, \ldots, Z_k = z_k] = \sum_{\substack{\{i,j\} \in E \\ i,j \leq k,\ z_i \neq z_j}} w_{ij} + \frac{1}{2} \sum_{\substack{\{i,j\} \in E \\ i > k \text{ or } j > k}} w_{ij}. \tag{3}$$

For the algorithm, we do not even need to evaluate this expression: only the difference between the two choices for the next vertex matters, and this difference involves only the edges to the vertices placed so far.

Algorithm (Greedy MAX-CUT):

1. Initialize $S \leftarrow \emptyset$ and $\bar{S} \leftarrow \emptyset$.
2. For $k \leftarrow 1$ to n: let $a_k \leftarrow \sum_{j \in S} w_{kj}$ and $b_k \leftarrow \sum_{j \in \bar{S}} w_{kj}$ be the total weight of edges from vertex k to the vertices already in S and in $\bar{S}$, respectively. If $a_k \geq b_k$, add k to $\bar{S}$; otherwise, add k to S.
3. Return the cut $(S, \bar{S})$.

Theorem 3.6. The greedy algorithm returns a cut of weight at least $W/2 \geq \mathrm{OPT}/2$ in time $O(n+m)$.

Proof. Let $z_1, \ldots, z_n$ be the values chosen by the algorithm, and let

$$E_k = \mathbb{E}[w(S) \mid Z_1 = z_1, \ldots, Z_k = z_k]$$

be the conditional expectation of the random cut weight given that the first k vertices are placed as the algorithm placed them. Then $E_0 = \mathbb{E}[w(S)] = W/2$ by Theorem 3.5, and E_n is the weight of the returned cut, since conditioning on all n values leaves nothing random. It therefore suffices to show $E_k \geq E_{k-1}$ for every k.

Fix $z_1, \ldots, z_{k-1}$. Since Z_k is uniform on $\{0,1\}$ and independent of $Z_1, \ldots, Z_{k-1}$, the law of total expectation gives

$$\begin{aligned} E_{k-1} = {} & \frac{1}{2}\,\mathbb{E}[w(S) \mid Z_1 = z_1, \ldots, Z_{k-1} = z_{k-1}, Z_k = 0] \\ & + \frac{1}{2}\,\mathbb{E}[w(S) \mid Z_1 = z_1, \ldots, Z_{k-1} = z_{k-1}, Z_k = 1]. \end{aligned}$$

An average of two numbers is at most their maximum, so if z_k is chosen to maximize the conditional expectation, then $E_k \geq E_{k-1}$. It remains to check that the algorithm makes this choice. By (3), the two conditional expectations differ only in the edges between k and $\{1, \ldots, k-1\}$: all other edges either do not involve k or have an endpoint beyond k, and contribute the same in both cases. Placing k in $\bar{S}$ cuts exactly the edges from k to S, of total weight a_k, and placing k in S cuts exactly those to $\bar{S}$, of total weight b_k. The algorithm chooses the larger of the two, which is the maximizing choice.

For the running time, each edge $\{j, k\}$ with $j < k$ is examined exactly once, when vertex k is placed. This concludes the proof. □

Alternatively, any locally optimal cut (where no single vertex move improves the cut) has weight at least $W/2$: local optimality means that each vertex has at least half its weighted degree on cut edges, and summing over all vertices gives $w(S) \geq W/2$.

We can formulate MAX-CUT as an ILP: for each edge $\{i, j\}$, introduce $y_{ij} \in \{0,1\}$ indicating whether it is cut, and for each vertex i, introduce $x_i \in \{0,1\}$ indicating its side. An edge is cut when its endpoints are on different sides:

$$\begin{aligned} \text{maximize} \quad & \sum_{\{i,j\} \in E} w_{ij}\, y_{ij} \\ \text{subject to} \quad & y_{ij} \leq x_i + x_j, \quad y_{ij} \leq 2 - x_i - x_j && \text{for all } \{i,j\} \in E \\ & x_i \in \{0,1\},\ y_{ij} \in \{0,1\} && \text{for all } i, \{i,j\} \end{aligned}$$

The LP relaxation replaces integrality with $0 \leq x_i, y_{ij} \leq 1$. Unfortunately, this relaxation is too weak: on any graph, setting $x_i = 1/2$ for all i and $y_{ij} = 1$ for all edges is feasible, with LP value W. Since $\mathrm{OPT} \leq W$, the LP always achieves $\mathrm{OPT}_{\mathrm{LP}} = W$. On the complete graph K_n with unit weights, $\mathrm{OPT} \approx n^2/4$ while $W = \binom{n}{2} \approx n^2/2$, so the integrality gap is $1/2$—no better than the trivial randomized algorithm. LP relaxation provides no advantage for MAX-CUT.

To break the 1/2 barrier, Goemans and Williamson [34] used a stronger relaxation based on *semidefinite programming* (SDP). The starting point is a different integer formulation, better suited to relaxation. Instead of $x_i \in \{0,1\}$, assign $z_i \in \{-1,+1\}$ to each vertex: $S = \{i : z_i = +1\}$. An edge $\{i,j\}$ is cut if and only if $z_i \neq z_j$, which happens when $z_i z_j = -1$. The objective becomes:

$$\text{maximize} \quad \sum_{\{i,j\} \in E} w_{ij} \cdot \frac{1 - z_i z_j}{2}$$

subject to $z_i \in \{-1,+1\}$ for all i.

The SDP relaxation replaces each scalar z_i with a unit vector $v_i \in \mathbb{R}^n$, and each product $z_i z_j$ with the inner product $v_i \cdot v_j$. The constraint $z_i^2 = 1$ becomes $\|v_i\| = 1$. The relaxation is:

$$\begin{aligned} \text{maximize} \quad & \sum_{\{i,j\} \in E} w_{ij} \cdot \frac{1 - v_i \cdot v_j}{2} \\ \text{subject to} \quad & \|v_i\| = 1 \qquad\qquad \text{for all } i \in V \end{aligned} \tag{4}$$

This is a relaxation: every $z \in \{-1,+1\}^n$ yields the feasible solution $v_i = z_i e_1$, where e_1 is the first standard basis vector, with the same objective value, since $v_i \cdot v_j = z_i z_j$. Hence $\text{OPT}_{\text{SDP}} \geq \text{OPT}$. The dimension n of the vectors is not a restriction either: n vectors always span a space of dimension at most n.

Optimizing over vectors looks like a nonlinear, nonconvex problem. The reason it is nonetheless tractable is that the objective and the constraints depend on the vectors only through their pairwise inner products $v_i \cdot v_j$, and the set of all possible inner product matrices has a clean description. To state it, we need some linear algebra.

A symmetric matrix $Y \in \mathbb{R}^{n \times n}$ is *positive semidefinite*, written $Y \succeq 0$, if ${}^t x Y x \geq 0$ for all $x \in \mathbb{R}^n$. For a symmetric matrix Y, the following are equivalent:

1. $Y \succeq 0$;
2. all eigenvalues of Y are nonnegative;
3. $Y = {}^t V V$ for some matrix $V \in \mathbb{R}^{d \times n}$, i.e., $Y_{ij} = v_i \cdot v_j$ is the *Gram matrix* of the columns $v_1, \ldots, v_n \in \mathbb{R}^d$ of V.

Indeed, if $Y = {}^t V V$, then ${}^t x Y x = {}^t(Vx)(Vx) = \|Vx\|^2 \geq 0$, so (3) implies (1). Testing (1) on an eigenvector x with eigenvalue λ gives $\lambda \|x\|^2 = {}^t x Y x \geq 0$, so (1) implies (2). Finally, a symmetric matrix has a spectral decomposition $Y = Q \Lambda {}^t Q$ with Q orthogonal and Λ the diagonal matrix of eigenvalues; if these are nonnegative, then $V = \Lambda^{1/2} {}^t Q$ satisfies ${}^t V V = Y$, so (2) implies (3). In practice, a factorization $Y = {}^t L L$ with L upper triangular is computed by the Cholesky decomposition in $O(n^3)$ time.

The positive semidefinite matrices form a convex cone: if $Y, Y' \succeq 0$ and $\lambda, \mu \geq 0$, then $\lambda Y + \mu Y' \succeq 0$, directly from the definition. A semidefinite program optimizes a linear function of the entries of Y over this cone intersected with finitely many linear equations.

Definition 3.3: Semidefinite program

A semidefinite program (SDP) is an optimization problem of the form

$$\begin{aligned} \text{maximize} \quad & \langle C, Y \rangle \\ \text{subject to} \quad & \langle A_k, Y \rangle = b_k \qquad\qquad \text{for } k = 1, \ldots, p \\ & Y \succeq 0 \end{aligned} \tag{5}$$

where $C, A_1, \ldots, A_p \in \mathbb{R}^{n \times n}$ are symmetric matrices, $b \in \mathbb{R}^p$, and $\langle A, Y \rangle = \sum_{i,j} A_{ij} Y_{ij} = \text{tr}({}^t A Y)$ is the entrywise inner product of matrices.

An SDP is thus a linear program whose variable is a matrix, with the constraint $x \geq 0$ on a vector replaced by the constraint $Y \succeq 0$ on a matrix. It is a genuine generalization of LP: if C and all A_k are diagonal, only the diagonal of Y matters, and a diagonal matrix is positive semidefinite if and only if its diagonal entries are nonnegative, so (5) is an LP in the variables $Y_{11}, \dots, Y_{nn}$. Inequality constraints $\langle A_k, Y \rangle \leq b_k$ can be added by introducing slack variables on the diagonal of an enlarged matrix. Unlike an LP, the feasible region of an SDP is not a polyhedron: for instance, $\left(\begin{smallmatrix} x & y \\ y & z \end{smallmatrix}\right) \succeq 0$ if and only if $x, z \geq 0$ and $xz \geq y^2$, a region with a curved boundary. It is, however, convex, and this is what makes SDPs tractable: for every $\varepsilon > 0$, a feasible solution of value at least $\mathrm{OPT}_{\mathrm{SDP}} - \varepsilon$ can be computed in time polynomial in the input size and in $\log(1/\varepsilon)$, for example by the ellipsoid method or by interior point methods. Exact solutions are out of reach in general, since the optimum of an SDP with rational data may be irrational. In the rounding algorithm below, the additive loss of ε is absorbed into the approximation ratio, and we ignore it from now on.

We can now write the vector program (4) as an SDP. Let Y be the Gram matrix of the vectors, $Y_{ij} = v_i \cdot v_j$. The constraint $\|v_i\| = 1$ becomes $Y_{ii} = 1$, and by the equivalence above, a symmetric matrix Y is the Gram matrix of some vectors if and only if $Y \succeq 0$. So the vector program is equivalent to:

$$\begin{aligned} \text{maximize} \quad & \sum_{\{i,j\} \in E} w_{ij} \cdot \frac{1 - Y_{ij}}{2} \\ \text{subject to} \quad & Y_{ii} = 1 \qquad\qquad \text{for all } i \in V \\ & Y \succeq 0 \end{aligned}$$

This is of the form (5): the constraint $Y_{ii} = 1$ is $\langle e_i \, {}^t e_i, Y \rangle = 1$, and using $Y_{ii} = Y_{jj} = 1$, the objective can be written as $\frac{1}{4} \sum_{\{i,j\} \in E} w_{ij}(Y_{ii} + Y_{jj} - 2Y_{ij}) = \frac{1}{4}\langle L_G, Y \rangle$, where $L_G = D - A_G$ is the weighted Laplacian of G, with A_G the weighted adjacency matrix and D the diagonal matrix of weighted degrees. To recover vectors for rounding from an optimal matrix Y^*, we compute the Cholesky decomposition $Y^* = {}^t L \, L$; the columns of L are unit vectors $v_1^*, \dots, v_n^*$ with $v_i^* \cdot v_j^* = Y_{ij}^*$.

Let $v_1^*, \dots, v_n^*$ be an optimal SDP solution with value $\mathrm{OPT}_{\mathrm{SDP}} \geq \mathrm{OPT}$. The rounding step is a single random choice: a hyperplane through the origin, which partitions the vertices.

Goemans–Williamson algorithm:

1. Solve the SDP relaxation (4) to obtain unit vectors $v_1^*, \dots, v_n^*$.
2. Choose a uniformly random unit vector r in the same space as the v_i^*.
3. Set $S \leftarrow \{i \in V : v_i^* \cdot r \geq 0\}$.
4. Return the cut $(S, \bar{S})$.

The analysis rests on a geometric lemma: the probability that a random hyperplane separates two unit vectors equals the angle between them divided by π.

Lemma 3.1. For unit vectors $v_i, v_j \in \mathbb{R}^d$ and a uniformly random unit vector $r \in \mathbb{R}^d$,

$$\mathbb{P}[\mathrm{sign}(v_i \cdot r) \neq \mathrm{sign}(v_j \cdot r)] = \frac{\arccos(v_i \cdot v_j)}{\pi}.$$

Proof. The random hyperplane $\{x : r \cdot x = 0\}$ separates v_i and v_j if and only if it passes between them. Restricting to the two-dimensional plane spanned by v_i and v_j, the hyperplane intersects this plane in a random line through the origin. If the angle between v_i and v_j is $\theta = \arccos(v_i \cdot v_j)$, the line separates them if and only if it falls in one of the two arcs of angle θ (see Figure 3.1), which happens with probability $2\theta/(2\pi) = \theta/\pi$. □

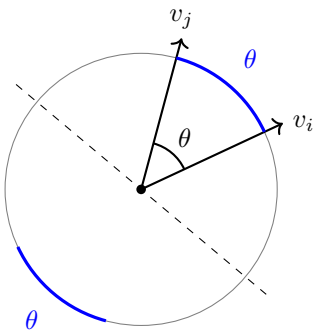


Figure 3.1: The random hyperplane (dashed line through the origin) separates v_i and v_j if and only if it passes through one of the two arcs of angle θ (shown in blue). This occurs with probability $2\theta/(2\pi) = \theta/\pi$.

Theorem 3.7. The algorithm above achieves

$$\mathbb{E}[w(S)] \geq \alpha \cdot \mathrm{OPT}, \quad \text{where } \alpha = \min_{0 \leq \theta \leq \pi} \frac{2}{\pi} \cdot \frac{\theta}{1 - \cos\theta} \geq 0.878.$$

Proof. By linearity of expectation and Lemma 3.1:

$$\mathbb{E}[w(S)] = \sum_{\{i,j\} \in E} w_{ij} \cdot \frac{\arccos(v_i^* \cdot v_j^*)}{\pi}$$

We compare this to the SDP value, which uses $(1 - v_i^* \cdot v_j^*)/2$ per edge. For each edge $\{i, j\}$, let $\theta_{ij} = \arccos(v_i^* \cdot v_j^*)$, so $v_i^* \cdot v_j^* = \cos\theta_{ij}$. We need to show $\frac{\theta}{\pi} \geq \alpha \cdot \frac{1-\cos\theta}{2}$ for all $\theta \in [0, \pi]$, i.e., $\frac{2\theta}{\pi(1-\cos\theta)} \geq \alpha$.

The function $g(\theta) = \frac{2\theta}{\pi(1-\cos\theta)}$ is continuous on $(0, \pi]$, with $g(\theta) \to \infty$ as $\theta \to 0^+$ and $g(\pi) = 1$. Since g is continuous and tends to $+\infty$ at 0, it attains its minimum α on $(0, \pi]$. Numerical computation gives $\alpha \approx 0.878$.

Therefore $\mathbb{E}[w(S)] \geq \alpha \sum_{\{i,j\} \in E} w_{ij} \cdot \frac{1 - v_i^* \cdot v_j^*}{2} = \alpha \cdot \mathrm{OPT}_{\mathrm{SDP}} \geq \alpha \cdot \mathrm{OPT}$. □

The improvement from 0.5 to 0.878 comes entirely from the stronger relaxation: the SDP bound $\mathrm{OPT}_{\mathrm{SDP}}$ is much tighter than the LP bound $\mathrm{OPT}_{\mathrm{LP}} = W$. The SDP relaxation has an integrality gap of exactly $\alpha \approx 0.878$.

The Goemans–Williamson result was the first to show that semidefinite programming gives strictly better approximation ratios than linear programming. Assuming the Unique Games Conjecture [4], the ratio $\alpha \approx 0.878$ is the best possible for MAX-CUT in polynomial time.

September 21, 2026

Lecture 4: Approximation Algorithms IV

This last lecture on approximation algorithms treats two topics. The first is clustering: given n points and a number k, group the points into k clusters so that each point is close to the center of its cluster. Clustering problems are among the most common optimization problems in practice, and the k-center problem admits a particularly clean greedy approximation with a matching hardness bound. The second topic is a general design paradigm we have not yet formalized: *local search*, which starts from an arbitrary solution and repeatedly applies small improving modifications. We analyze local search on two problems: MAX-CUT, where it derandomizes the random cut of Lecture 3, and k-median, where a careful accounting of swaps yields a constant-factor approximation.

4.1 k-Center

In the *k-center* problem, we are given the metric space (X, d) of $|X| = n$ points and an integer $k \geq 1$, and we must choose a set $S \subseteq X$ of at most k *centers* minimizing the covering radius

$$r(S) = \max_{x \in X} d(x, S) \ .$$

Equivalently, we look for the smallest radius r such that k balls of radius r cover all of X.

The following greedy algorithm is due to Gonzalez [13] and is often called *farthest-first traversal*.

Algorithm: Farthest-First Traversal

1. Pick an arbitrary point $c_1 \in X$.
2. For $i \leftarrow 2$ to k: pick $c_i \in X$ maximizing $d\big(c_i, \{c_1, \dots, c_{i-1}\}\big)$.
3. Return $S \leftarrow \{c_1, \dots, c_k\}$.

The algorithm greedily attacks the currently worst-served point. Its analysis rests on a simple dichotomy: either the returned radius is already small, or the algorithm has found $k+1$ points that are pairwise far apart, which is impossible if a good solution exists.

Theorem 4.1. Farthest-first traversal is a 2-approximation algorithm for k-center and runs in $O(nk)$ time.

Proof. Let $r = r(S)$ be the radius of the returned solution, attained by some point v with $d(v, S) = r$, and let r^* denote the optimal radius. Consider the $k+1$ points $c_1, \dots, c_k, v$. We claim that they are pairwise at distance at least r. For $j < i \leq k$:

$$d(c_i, c_j) \ \geq \ d\big(c_i, \{c_1, \dots, c_{i-1}\}\big) \ \geq \ d\big(v, \{c_1, \dots, c_{i-1}\}\big) \ \geq \ d(v, S) \ = \ r \ ,$$

where the second inequality holds because c_i was chosen to maximize the distance to the first $i-1$ centers, and the third because distances to a set can only decrease when the set grows. Moreover, $d(v, c_i) \geq d(v, S) = r$ for every i.

Now fix an optimal solution with centers $s_1^*, \dots, s_k^*$; its balls of radius r^* cover X. By the pigeonhole principle, two of our $k+1$ points, say x and y, lie in the ball of the same optimal center s^*. By the triangle inequality,

$$r \ \leq \ d(x, y) \ \leq \ d(x, s^*) + d(s^*, y) \ \leq \ 2r^* \ .$$

For the running time, maintain for every point its distance to the current center set; each of the k

iterations updates these n values with one distance evaluation each. □

The analysis is tight already for $k = 1$. Let $X = \{0, 1, \ldots, 2r\}$ with $d(i, j) = |i-j|$, the vertices of a path with unit edge lengths. The optimal center is the midpoint r, with radius $r^* = r$, whereas the algorithm may pick the endpoint $c_1 = 0$, with radius $2r$. For general k, place k copies of this path at mutual distance much larger than r: the algorithm can pick an endpoint in each copy, since after choosing one center in a copy, the farthest points lie in the copies not yet visited.

Hsu and Nemhauser [14] showed that the factor 2 is best possible.

Theorem 4.2. For every $\varepsilon > 0$, there is no polynomial-time $(2 - \varepsilon)$-approximation algorithm for k-center unless P = NP.

Proof. We reduce from the NP-complete *dominating set* problem: given a graph $G = (V, E)$ and an integer k, decide whether there is a set $D \subseteq V$ with $|D| \leq k$ such that every vertex is in D or adjacent to a vertex of D.

Given (G, k), define a metric on $X = V$ by $d(u, v) = 1$ if $\{u, v\} \in E$ and $d(u, v) = 2$ if $u \neq v$ and $\{u, v\} \notin E$. The triangle inequality holds because all distances are 1 or 2. If G has a dominating set D of size at most k, then $r(D) \leq 1$, so the optimal k-center radius is 1. If G has no such dominating set, then every set of k centers leaves some point at distance 2, so the optimal radius is 2.

A $(2-\varepsilon)$-approximation algorithm returns, in the first case, a solution of radius at most $(2-\varepsilon) \cdot 1 < 2$, hence of radius 1, which is a dominating set of size at most k. In the second case, every solution has radius 2. Comparing the returned radius to 2 therefore decides dominating set in polynomial time. □

4.2 Local Search

A *local search* algorithm is specified by a set of feasible solutions and, for each solution, a *neighborhood* of solutions reachable by a small modification. The algorithm starts at an arbitrary feasible solution and repeatedly moves to a neighbor of better objective value, until no neighbor improves on the current solution; such a solution is called a *local optimum*. Two questions decide whether this is a good algorithm: how good are local optima compared to global ones, and how many improving moves can occur?

We first revisit MAX-CUT in the notation of Lecture 3: given an undirected graph $G = (V, E)$ with edge weights $w_{ij} \geq 0$ and total weight W, find a cut $(S, \bar{S})$ maximizing the weight $w(S)$ of the crossing edges. In Lecture 3, we showed that a uniformly random cut has expected weight at least $W/2$ (Theorem 3.5), derandomized this by conditional expectations, and remarked that every locally optimal cut has weight at least $W/2$ as well. We now prove this remark as a first instance of the paradigm, and, more importantly for local search, bound the number of improving moves. Take as neighborhood the *flip* of a single vertex to the other side.

Theorem 4.3. Every local optimum $(S, \bar{S})$ of flip-based local search for MAX-CUT satisfies $w(S) \geq W/2$. On unweighted graphs, a local optimum is reached after at most $|E|$ flips.

Proof. Let $(S, \bar{S})$ be a local optimum. For a vertex v, let $\mathrm{cut}(v)$ and $\mathrm{uncut}(v)$ denote the total weight of the edges at v that do and do not cross the cut. Flipping v changes the cut weight by $\mathrm{uncut}(v) - \mathrm{cut}(v)$, so local optimality means $\mathrm{cut}(v) \geq \mathrm{uncut}(v)$ for every v. Summing over all vertices counts every edge

twice:

$$2\,w(S) = \sum_{v \in V} \mathrm{cut}(v) \geq \sum_{v \in V} \mathrm{uncut}(v) = 2\,\big(W - w(S)\big) \ ,$$

so $w(S) \geq W/2$. On unweighted graphs, every flip increases the number of cut edges by at least 1, and this number is bounded by $|E|$, so at most $|E|$ flips occur. □

The guarantee of Theorem 4.3 concerns the quality of *every* local optimum; nothing needed to be known about how the local search reaches one. This separation of concerns is typical for local search analyses, including the one we do next.

The running-time argument, on the other hand, breaks down for weighted graphs: each flip improves the cut weight, but possibly by a tiny amount, and there are weighted instances on which local search performs an exponential number of flips. The standard remedy is to accept a move only if it improves the objective by a fixed factor. Then the objective moves geometrically, and the number of moves is logarithmic in the ratio of the largest to the smallest possible objective value.

Lemma 4.1. Let $0 < \delta \leq 1$. Consider a local search that moves from S to a neighbor S' only if $\mathrm{cost}(S') \leq (1-\delta)\,\mathrm{cost}(S)$, for a minimization problem, or only if $w(S') \geq (1+\delta)\,w(S)$, for a maximization problem. If the objective is at most C initially and every nonzero objective value is at least c, then the number of moves is at most $2\ln(C/c)/\delta + 1$.

Proof. For minimization, after t moves the cost is at most $(1-\delta)^t C \leq e^{-\delta t} C$. If $t > \ln(C/c)/\delta$, this is less than c, so the cost is zero and no further move is possible. For maximization, start the count at the first solution with nonzero objective, at least c; after t moves the objective is at least $(1+\delta)^t c$ and at most C, so $t \leq \ln(C/c)/\ln(1+\delta) \leq 2\ln(C/c)/\delta$, using $\ln(1+\delta) \geq \delta/2$ for $\delta \leq 1$. □

For MAX-CUT, start from a cut separating the endpoints of a heaviest edge, so that $c = w_{\max}$ and $C = W \leq |E|\,w_{\max}$, and flip only if the cut weight grows by a factor at least $1+\delta$ with $\delta = \varepsilon/n$. By Lemma 4.1, at most $O\big(\frac{n}{\varepsilon}\log|E|\big)$ flips occur. The price is paid in the guarantee. At the end, $\mathrm{uncut}(v) - \mathrm{cut}(v) \leq \delta\,w(S)$ for every v, and summing over v as in the proof of Theorem 4.3 gives $2W - 4w(S) \leq n\delta\,w(S) = \varepsilon\,w(S)$, that is,

$$w(S) \ \geq \ \frac{W}{2} \cdot \frac{1}{1+\varepsilon/4} \ \geq \ \Big(\frac{1}{2} - \frac{\varepsilon}{8}\Big) W \ .$$

The same device will make the k-median local search below run in polynomial time.

4.3 k-Median

The *k-median* problem is the sum-version of k-center: choose $S \subseteq X$ with $|S| = k$ minimizing

$$\mathrm{cost}(S) = \sum_{x \in X} d(x, S) \ .$$

Compared to k-center, the objective is far more robust: a single outlier point moves the k-center optimum arbitrarily, but shifts the k-median objective by only its own distance. The price of robustness is that no analog of the clean greedy algorithm of Theorem 4.1 is known; the natural algorithms for k-median are based on linear programming or on local search. We analyze the simplest local search, due to Arya, Garg, Khandekar, Meyerson, Munagala, and Pandit [16], with the streamlined proof of Gupta and Tangwongsan.

The neighborhood is given by *swaps*: from a solution S, remove one center $s \in S$ and add one non-center $o \notin S$.

Algorithm (Local search with swaps):

1. Start with an arbitrary set $S \subseteq X$ of k centers.
2. While there are $s \in S$ and $o \notin S$ with $\mathrm{cost}(S - s + o) < \mathrm{cost}(S)$: replace S by $S - s + o$.
3. Return S.

The algorithm returns a solution that no single swap can improve. We call S *swap-optimal* if

$$\mathrm{cost}\big(S - s + o\big) \geq \mathrm{cost}(S) \qquad \text{for all } s \in S,\ o \notin S\ ,$$

where $S - s + o$ abbreviates $(S \setminus \{s\}) \cup \{o\}$.

Fix a swap-optimal solution S and an optimal solution O, both of size k. For a point x, let $d_S(x) = d(x, S)$ and $d_O(x) = d(x, O)$ be its distances to the two solutions, let $s(x) \in S$ and $o(x) \in O$ be centers attaining them, and let $N_S(s) = \{x : s(x) = s\}$ and $N_O(o) = \{x : o(x) = o\}$ be the clusters of S and O.

The proof plan is this: for suitable pairs (s, o), swap-optimality gives an inequality $0 \leq \mathrm{cost}(S - s + o) - \mathrm{cost}(S)$, and we upper-bound the right-hand side by exhibiting an explicit assignment of the points to the centers of $S - s + o$. Summing these inequalities over a well-chosen set of k swaps will produce $\mathrm{cost}(S) \leq 5 \cdot \mathrm{cost}(O)$.

Which assignment? The points of $N_O(o)$ are easy: send them to the new center o, at their optimal cost $d_O(x)$. Points that keep their center cost nothing extra. The problem is the points of $N_S(s)$ that are not in $N_O(o)$: they lose their center s, and we need a new center of S for them that is provably not much farther than s was. The optimal center $o(x)$ of such a point is close to x, but it is not available, since it is not in S. What is available is the center $s(o(x))$ of S closest to $o(x)$, and the detour $x \to o(x) \to s(o(x))$ can be charged to the optimal solution: the first leg costs $d_O(x)$, and the second leg is at most $d(o(x), s(x))$, since $s(x)$ is one of the candidates for $s(o(x))$, which in turn is at most $d_O(x) + d_S(x)$ by the triangle inequality. The next lemma records this.

Lemma 4.2. Recall that $s(o)$, for $o \in O$, is a center of S closest to o, since o is itself a point of X. For every point x,

$$d\big(x, s(o(x))\big) \ \leq\ 2d_O(x) + d_S(x)\ .$$

Proof. By the triangle inequality and the choice of $s(o(x))$ as a closest center,

$$d\big(o(x), s(o(x))\big) \ \leq\ d\big(o(x), s(x)\big) \ \leq\ d\big(o(x), x\big) + d\big(x, s(x)\big) \ =\ d_O(x) + d_S(x)\ .$$

Hence $d\big(x, s(o(x))\big) \leq d\big(x, o(x)\big) + d\big(o(x), s(o(x))\big) \leq d_O(x) + (d_O(x) + d_S(x))$. □

So each rerouted point costs at most $2d_O(x)$ more than before, an amount the optimal solution can pay for. This leaves the choice of the swaps. The obvious candidates are the k swaps $(s(o), o)$ for $o \in O$: each optimal center enters once, replacing the center of S that stands in for it. Two things can go wrong. First, the reroute of a point $x \in N_S(s)$ presupposes that $s(o(x)) \neq s$; if several optimal centers o, o' share the same $s(o) = s(o') = s$, then in the swap (s, o) the points of $N_S(s)$ that belong to $N_O(o')$ would be sent to $s(o') = s$, the very center being removed. Second, if the same center s is swapped out in many pairs, the charge $2d_O(x)$ on its cluster $N_S(s)$ is collected many times over, and the sum of the inequalities no longer compares to $\mathrm{cost}(O)$. The following lemma shows that both problems can be avoided at once, at the price of letting a center of S appear in two pairs rather than one. If $o \mapsto s(o)$ happens to be a bijection from O onto S, the pairs $(s(o), o)$ satisfy all three conditions with every s appearing once, and the computation below gives $\mathrm{cost}(S) \leq 3\,\mathrm{cost}(O)$; the factor 5 is the price of the general case.

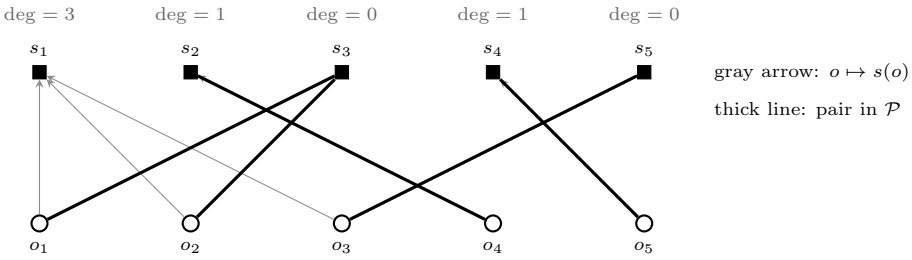


Figure 4.1: The pairing of Lemma 4.3 for $k = 5$. Squares are the centers of S, circles those of O, and an arrow goes from each o to $s(o)$, so the degree of a center of S is the number of arrows into it; here $n_0 = 2$, $n_1 = 2$, $n_2 = 1$. The centers s_2 and s_4 of degree 1 are paired with their unique preimages. The $k - n_1 = 3$ remaining optimal centers o_1, o_2, o_3, all mapped to s_1, are matched to the $n_0 = 2$ centers of degree 0, at most two each. No paired center of S receives an arrow from an optimal center other than its partner, which is condition 3, and s_1 is never swapped out.

Lemma 4.3. There is a set $\mathcal{P}$ of k swap pairs $(s, o) \in S \times O$ such that:

1. every $o \in O$ appears in exactly one pair;
2. every $s \in S$ appears in at most two pairs;
3. if $(s, o) \in \mathcal{P}$, then $s(o') \neq s$ for every $o' \in O \setminus \{o\}$.

Proof. For $s \in S$, let $\deg(s) = |\{o \in O : s(o) = s\}|$ be the number of optimal centers whose closest center of S is s, and let n_0, n_1, and n_2 be the numbers of centers $s \in S$ with $\deg(s) = 0$, with $\deg(s) = 1$, and with $\deg(s) \geq 2$, respectively. Pair every s with $\deg(s) = 1$ with the unique o such that $s(o) = s$; this creates n_1 pairs satisfying condition 3 (Figure 4.1 shows an example). The remaining $k - n_1$ optimal centers are matched to centers of degree 0, at most two per center; condition 3 then holds trivially. This is possible because $n_0 + n_1 + n_2 = k$ and

$$k \;=\; \sum_{s \in S} \deg(s) \;\geq\; n_1 + 2n_2 \;,$$

whence $n_2 \leq (k - n_1)/2$ and thus $n_0 = k - n_1 - n_2 \geq (k - n_1)/2$. □

Theorem 4.4. Every swap-optimal solution S satisfies $\text{cost}(S) \leq 5 \cdot \text{cost}(O)$.

Proof. Fix a pair $(s, o) \in \mathcal{P}$ from Lemma 4.3. Then $0 \leq \text{cost}(S - s + o) - \text{cost}(S)$: by swap-optimality if $o \notin S$, and trivially if $o \in S$, since then $S - s + o \subseteq S$ and removing centers cannot decrease the cost. We bound $\text{cost}(S - s + o)$ from above by assigning every point to some center of $S - s + o$ (an upper bound, since cost uses the closest center):

- every $x \in N_O(o)$ is assigned to o, at cost $d_O(x)$;
- every $x \in N_S(s) \setminus N_O(o)$ is assigned to $s(o(x))$, at cost at most $2d_O(x) + d_S(x)$ by Lemma 4.2; this center is available: $o(x) \neq o$ because $x \notin N_O(o)$, hence $s(o(x)) \neq s$ by condition 3 of

Lemma 4.3;

- every other point x keeps its center $s(x) \neq s$, at cost $d_S(x)$.

Subtracting $\text{cost}(S) = \sum_x d_S(x)$, the first group contributes $d_O(x) - d_S(x)$ per point, the second at most $(2d_O(x) + d_S(x)) - d_S(x) = 2d_O(x)$, and the third nothing:

$$0 \;\le\; \text{cost}(S - s + o) - \text{cost}(S) \;\le\; \sum_{x \in N_O(o)} (d_O(x) - d_S(x)) \;+\; \sum_{x \in N_S(s)} 2d_O(x) \;,$$

where we extended the second sum from $N_S(s) \setminus N_O(o)$ to all of $N_S(s)$, which is safe because $2d_O(x) \geq 0$.

Now sum this inequality over all k pairs of $\mathcal{P}$. By condition 1, the clusters $N_O(o)$ appear exactly once each and partition X, so the first sums add up to $\text{cost}(O) - \text{cost}(S)$. By condition 2, each cluster $N_S(s)$ appears at most twice, and the clusters partition X, so the second sums add up to at most $2\sum_{x \in X} 2d_O(x) = 4\,\text{cost}(O)$. Altogether $0 \leq \text{cost}(O) - \text{cost}(S) + 4\,\text{cost}(O)$, that is, $\text{cost}(S) \leq 5\,\text{cost}(O)$. □

Three concluding remarks. First, the factor 5 is tight for single swaps, but swapping p centers simultaneously improves the guarantee to $3 + 2/p$ [16]; the proof follows the same pattern with a more careful pairing. Second, as for MAX-CUT, the number of improving swaps need not be polynomial, and the remedy is the same. Accept a swap only if it decreases the cost by a factor at least $1 - \delta$ with $\delta = \varepsilon/(2k)$, for some $0 < \varepsilon \leq 1$. With integer distances at most Δ, the initial cost is at most $n\Delta$ and every nonzero cost is at least 1, so Lemma 4.1 bounds the number of swaps by $O\big(\frac{k}{\varepsilon} \log(n\Delta)\big)$, polynomial in the input size and $1/\varepsilon$. At the end, each swap inequality holds up to an additive $\delta\,\text{cost}(S)$, so summing over the k pairs of $\mathcal{P}$ gives $0 \leq 5\,\text{cost}(O) - \text{cost}(S) + k\delta\,\text{cost}(S)$, that is,

$$\text{cost}(S) \;\le\; \frac{5}{1 - \varepsilon/2}\,\text{cost}(O) \;\le\; 5(1 + \varepsilon)\,\text{cost}(O) \;.$$

Third, the analysis uses only swap-optimality, not how it was reached, so the bound holds for the output of any procedure that stops at a swap-optimal solution.

September 28, 2026

Lecture 5: String Algorithms I

We have a long text and a short pattern, and we want to find every place the pattern appears in the text. The obvious method is to slide the pattern along and compare character by character. This takes $\Theta(nm)$ time, where n is the text length and m the pattern length. Any algorithm must read every character of the text and the pattern at least once, so $\Omega(n+m)$ is a lower bound. Can we match it with an $O(n+m)$ algorithm? The answer is yes, and the trick is the same in all algorithms we will meet: when a comparison fails, the characters we have already read tell us something about where the next match could start.

This lecture and the two following ones present a sequence of increasingly refined string-matching algorithms. In this lecture, we study the Morris–Pratt algorithm and its refinement by Knuth, which achieve $O(n+m)$ time by precomputing a failure function using $O(m)$ space. The next lecture treats the string-matching automaton of Simon and the Boyer–Moore algorithm; the last one presents the two-way algorithm of Crochemore and Perrin, which reduces the extra space to $O(1)$.

Given a string S, we write $S[k]$ for its kth character, and $S[i..j]$ for the substring from position i to j included. Given a text $T[1..n]$ and a pattern $P[1..m]$, the *string matching problem* asks to find all positions $i \in \{1, \dots, n-m+1\}$ such that $T[i..i+m-1] = P$. Each such i is an *occurrence* of P in T.

5.1 Morris–Pratt

Suppose we have just found a complete occurrence of P in the text. We need to keep searching, but do we really have to start over from scratch? Look at the occurrence we just matched: its tail end is still sitting in the text, and if that tail happens to equal the *beginning* of P, then we already have a head start on the next match. We just slide the pattern forward so that the matching prefix–suffix lines up, and resume comparing from where we left off. We never re-read a single text character (see Figure 5.1). This is the central idea of the Morris–Pratt (MP) algorithm [17].

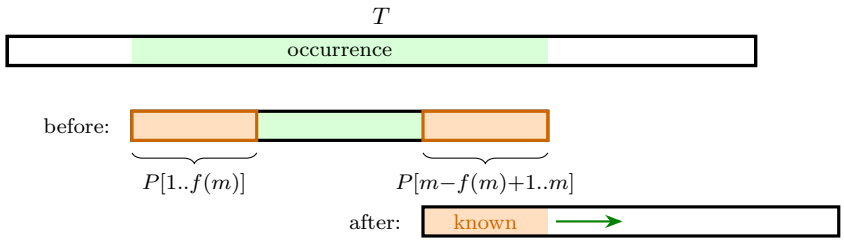


Figure 5.1: After a full match ($q = m$), MP sets $q = f(m)$. The orange suffix of the matched pattern equals the orange prefix by definition of f. These $f(m)$ characters already match the text at the shifted position, so the algorithm resumes at $P[f(m)+1]$ without re-reading.

The same idea works not only after a full match, but after any partial match. A proper prefix of a string that is also a suffix of it is called a *border* of the string. If we have matched the first q characters of P and then hit a mismatch, the longest border of $P[1..q]$ tells us exactly how far to shift. MP precomputes this for every q in the *failure function* $f\colon \{1, \dots, m\} \to \{0, \dots, m-1\}$:

$$f(q) = \max\{k : 0 \le k < q \text{ and } P[1..k] \text{ is a suffix of } P[1..q]\},$$

that is, $f(q)$ is the length of the longest border of $P[1..q]$.

Example. For $P =$ `ababac`:

q	1	2	3	4	5	6
$P[1..q]$	`a`	`ab`	`aba`	`abab`	`ababa`	`ababac`
$f(q)$	0	0	1	2	3	0

For instance, $f(5) = 3$ because the longest border of `ababa` is `aba`, which has length 3. Similarly, $f(4) = 2$ because `ab` is both a proper prefix and a suffix of `abab`.

How do we compute f? The trick is to match the pattern against *itself*. To see why this works, define the *failure chain* at q as the sequence $f(q), f(f(q)), \ldots, 0$ obtained by iterating f. This chain lists *every* border of $P[1..q]$, in decreasing order of length (see Figure 5.2). The reason is simple: a border of a border is itself a border, so the chain cannot skip any.

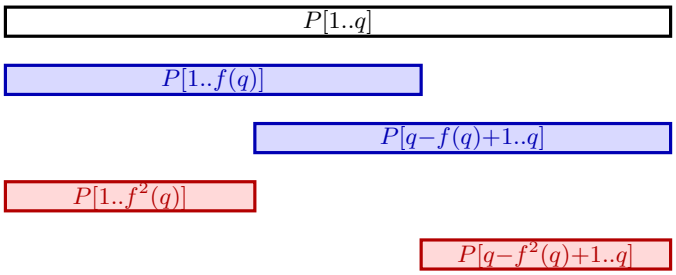


Figure 5.2: The failure chain at q. Each row shows a border of $P[1..q]$: the left-aligned bar is the prefix, the right-aligned bar is the matching suffix. A prefix of a prefix is a prefix, and a suffix of a suffix is a suffix, so every element of the chain $f(q), f^2(q), \ldots$ gives a border of $P[1..q]$.

Lemma 5.1. For $q \geq 2$:

$$f(q) = \begin{cases} k+1 & \text{if } k \text{ is the largest element of the failure chain at } q-1 \text{ such that } P[k+1] = P[q], \\ 0 & \text{if no such } k \text{ exists.} \end{cases}$$

Proof. Suppose $P[1..k{+}1]$ is a border of $P[1..q]$, with $k+1 \geq 1$. Then $P[k{+}1] = P[q]$ and $P[1..k]$ is a border of $P[1..q{-}1]$. We claim k appears in the failure chain at $q{-}1$. Indeed, $f(q{-}1) \geq k$ since $f(q{-}1)$ is the longest border; if $f(q{-}1) > k$, then $k \leq f(f(q{-}1))$ by the same reasoning applied to $P[1..f(q{-}1)]$, and by induction k is reached by iterating f. Conversely, any k in the chain with $P[k{+}1] = P[q]$ gives $P[1..k{+}1]$ as a border of $P[1..q]$. Taking the largest such k gives $f(q)$. □

In other words: to compute $f(q)$, we already know $f(q{-}1)$ from the previous iteration. We walk down the failure chain at $q{-}1$, testing whether $P[k{+}1] = P[q]$ at each step. The first k that passes gives $f(q) = k+1$; if none passes, $f(q) = 0$. This reduces the preprocessing to a single left-to-right pass over P, using the already computed values of f to guide each step. The resulting procedure takes $O(m)$ time in total, as we verify below after stating it.

Algorithm: Compute Failure Function

1. Set $f(1) \leftarrow 0$ and $k \leftarrow 0$.
2. For $q \leftarrow 2$ to m:
   (a) While $k > 0$ and $P[k+1] \neq P[q]$, set $k \leftarrow f(k)$.
   (b) If $P[k+1] = P[q]$, set $k \leftarrow k+1$.
   (c) Set $f(q) \leftarrow k$.

Lemma 5.2. The algorithm above correctly computes the failure function in $O(m)$ time.

Proof. We show that at the end of each outer iteration for index q, $f(q)$ is correctly set to the length of the longest border of $P[1..q]$. The invariant is that at the start of the iteration, $k = f(q-1)$. By Lemma 5.1, $f(q)$ is $k'+1$ where k' is the largest element of the failure chain at $q-1$ with $P[k'+1] = P[q]$, or 0 if none exists. The while loop walks down this chain, trying each k in decreasing order and stopping at the first match. If $P[k+1] = P[q]$, the algorithm sets $k = k+1$ and then $f(q) = k$, matching the formula. If no match is found, $k = 0$ and $f(q) = 0$. In both cases, $k = f(q)$ at the end of the iteration, maintaining the invariant for $q+1$.

For the running time: each of the $m-1$ outer iterations increments k by at most 1, so the total number of increments is at most $m-1$. Each while-loop iteration decreases k by at least 1 (since $f(k) < k$), and $k \geq 0$ always, so the total number of decrements is also at most $m-1$. Hence the while loop runs $O(m)$ times overall, giving $O(m)$ total time. □

The search phase is almost identical: we scan the text left to right, maintaining the length q of the current partial match. When a mismatch occurs, we walk down the failure chain, exactly as during preprocessing, until we find a border that can be extended, or reach 0.

Algorithm: MP Search

1. Compute the failure function f for P.
2. Set $q \leftarrow 0$.
3. For $i \leftarrow 1$ to n:
   (a) While $q > 0$ and $P[q+1] \neq T[i]$, set $q \leftarrow f(q)$.
   (b) If $P[q+1] = T[i]$, set $q \leftarrow q+1$.
   (c) If $q = m$, report an occurrence at position $i - m + 1$ and set $q \leftarrow f(q)$.

Theorem 5.1. The MP algorithm finds all occurrences of P in T in $O(n+m)$ time.

Proof. We maintain the invariant that after processing $T[i]$, q equals the length of the longest prefix of P that is a suffix of $T[1..i]$. Initially $q = 0$ and the invariant holds vacuously. At step i, if $P[q+1] = T[i]$, extending the match to $q+1$ is correct. If $P[q+1] \neq T[i]$, setting $q = f(q)$ is safe: any prefix of P that is a suffix of $T[1..i-1]$ and could extend to match $T[i]$ must also be a suffix of $P[1..q]$, hence its length appears in the failure chain at q. The while loop tries each candidate in decreasing order. Thus the invariant is preserved, and occurrences (when $q = m$) are reported correctly.

Preprocessing takes $O(m)$ time by Lemma 5.2. The search phase uses the same amortized argument: q increases by at most 1 per outer iteration (at most n increments total), and each while-loop

iteration decreases q by at least 1 with $q \geq 0$ always, so at most n decrements total. The search takes $O(n)$ time, giving $O(n+m)$ overall. □

Example. Let $P = \texttt{ababac}$, with failure function $f = (0, 0, 1, 2, 3, 0)$ as computed above, and let $T = \texttt{abababac}$. The table lists, for each text position i, the comparisons made against $T[i]$ and the value of q afterwards.

i	$T[i]$	comparisons against $T[i]$	q
1	a	$P[1] = \texttt{a}$	1
2	b	$P[2] = \texttt{b}$	2
3	a	$P[3] = \texttt{a}$	3
4	b	$P[4] = \texttt{b}$	4
5	a	$P[5] = \texttt{a}$	5
6	b	$P[6] = \texttt{c} \neq \texttt{b}$, so $q \leftarrow f(5) = 3$; then $P[4] = \texttt{b}$	4
7	a	$P[5] = \texttt{a}$	5
8	c	$P[6] = \texttt{c}$; occurrence at $8 - 6 + 1 = 3$; $q \leftarrow f(6) = 0$	0

At $i = 6$, the algorithm has matched ababa and reads a b where the pattern has a c. It does not move back in the text: the border aba of ababa is still matched, and one comparison later, so is abab.

5.2 Knuth–Morris–Pratt

The bound of Theorem 5.1 is amortized: *on average* over the whole text, MP spends $O(1)$ time per text character. The *delay*, that is, the number of pattern characters compared against a single text character, can nevertheless be large. Consider $P = \texttt{a}^m$ and a text character $T[i] = \texttt{b}$ arriving after a partial match of length $q = m - 1$. The failure chain at q is $m-2, m-3, \ldots, 0$, and MP compares $P[k+1] = \texttt{a}$ against $T[i] = \texttt{b}$ for every single k on it: $m - 1$ comparisons, all of them with the same predictable outcome.

The wasted work has a clear structure. When the comparison of $P[q+1]$ against $T[i]$ fails and we fall back to the border $k = f(q)$, the next comparison pits $P[k+1]$ against the same $T[i]$. If $P[k+1] = P[q+1]$, this comparison is doomed: it must fail for the same reason the previous one did. The refinement of Knuth [18] skips these doomed candidates ahead of time. Call a border $P[1..k]$ of $P[1..q]$ *strict* if $P[k+1] \neq P[q+1]$, that is, if the character following the border differs from the character whose comparison just failed. The *strict failure function* is

$$f'(q) = \max\big(\{k : P[1..k] \text{ is a strict border of } P[1..q]\} \cup \{0\}\big) \qquad \text{for } 1 \leq q < m,$$

and $f'(m) = f(m)$, since there is no character $P[m+1]$ to compare. The KMP search algorithm is the MP search algorithm with f replaced by f'.

Example. For $P = \texttt{ababac}$, compare the two failure functions:

q	1	2	3	4	5	6
$f(q)$	0	0	1	2	3	0
$f'(q)$	0	0	0	0	3	0

Take $q = 4$: the longest border of $\texttt{abab}$ is $\texttt{ab}$, but $P[3] = \texttt{a} = P[5]$, so after a mismatch of $P[5]$ the candidate is doomed and f' skips it; the next border is the empty one. At $q = 5$ the border $\texttt{aba}$ survives because $P[4] = \texttt{b} \neq \texttt{c} = P[6]$.

Lemma 5.3. The KMP search algorithm reports exactly the occurrences of P in T, in $O(n+m)$ time.

Proof. We show that the KMP and MP searches end every outer iteration with the same value of q; the claim then follows from Theorem 5.1. Fix a text position i and let q be the common value at the start of the iteration. Let j be the largest element of the failure chain at q with $P[j+1] = T[i]$ (with $j = q$ itself allowed), if one exists. The MP while loop descends the chain and stops exactly at j. Now consider the KMP while loop at some current chain element $k > j$, so that $P[k+1] \neq T[i]$. The jump to $f'(k)$ skips precisely the chain elements j' with $P[j'+1] = P[k+1]$; all of them satisfy $P[j'+1] \neq T[i]$, so j is never skipped. Moreover, $P[1..j]$ is a strict border of $P[1..k]$ because $P[j+1] = T[i] \neq P[k+1]$, so $f'(k) \geq j$. Hence the KMP iteration descends through chain elements that all fail against $T[i]$, never passes j, and cannot stop above j by maximality of j; it therefore stops at j as well, and both algorithms continue with $q = j+1$. If no such j exists, both loops reach 0 and treat $T[i]$ identically. The running-time analysis of Theorem 5.1 applies verbatim, since $f'(q) \leq f(q) < q$. □

The point of the modification is the delay. By definition of strictness, two *consecutive* candidates tried against the same text character now always differ: if the comparison $P[q+1] \neq T[i]$ fails and we fall back to $k = f'(q) > 0$, then $P[k+1] \neq P[q+1]$, so the next comparison is not a repetition of the previous one. On $P = \texttt{a}^m$, the strict failure function is identically 0, and the delay drops from $m-1$ to 1. In general, the delay is logarithmic [18]. The proof needs a second way of looking at borders. An integer $p \geq 1$ is a *period* of a string $S[1..k]$ if $S[i] = S[i+p]$ for all $1 \leq i \leq k-p$; every $p \geq k$ is a period vacuously, and $\mathrm{per}(S)$ denotes the smallest period. Borders and periods are two descriptions of the same thing: S has a border of length b if and only if it has period $k-b$, since both statements say that $S[i] = S[i+k-b]$ for $1 \leq i \leq b$. In particular, the failure chain at k lists the periods of $P[1..k]$ smaller than k, and $\mathrm{per}(P[1..k]) = k - f(k)$. A string may have several periods, but they cannot be independent.

Lemma 5.4 (Periodicity lemma). If p and q are periods of a string S of length at least $p+q$, then so is $\gcd(p,q)$.

Proof. By induction on $p+q$. The case $p = q$ is trivial, so let $p > q$. We show that $p-q$ is a period of S. Let $1 \leq i \leq |S| - (p-q)$. If $i+p \leq |S|$, then $S[i] = S[i+p] = S[i+p-q]$, using period p and then period q backwards. Otherwise $i > |S| - p \geq q$, and $S[i] = S[i-q] = S[i-q+p]$, using period q backwards and then period p. Now q and $p-q$ are periods of S with $q + (p-q) = p \leq |S|$ and $\gcd(q, p-q) = \gcd(p,q)$, so the induction hypothesis applies. □

The lemma holds under the weaker hypothesis $|S| \geq p+q-\gcd(p,q)$, which is tight; this is the theorem of Fine and Wilf [19]. The version above is all we need.

Lemma 5.5. Let $1 \leq a < m$, $b = f'(a)$, and $c = f'(b)$. If $c \geq 1$, then $a > b + c$.

Proof. Since $c \geq 1$, it is a strict border of $P[1..b]$: $P[1..c]$ is a border of $P[1..b]$ and $P[c+1] \neq P[b+1]$. A border of a border is a border, so $P[1..b]$ and $P[1..c]$ are both borders of $P[1..a]$, which therefore has the periods $a-b$ and $a-c$. Suppose $a \leq b+c$, that is, $a \geq (a-b)+(a-c)$. By Lemma 5.4, $g = \gcd(a-b, a-c)$ is a period of $P[1..a]$, and g divides $(a-c)-(a-b) = b-c$. Stepping from position $c+1$ to position $b+1$ in increments of g stays inside $P[1..a]$, because $b+1 \leq a$, and yields $P[c+1] = P[b+1]$, contradicting the strictness of c. □

Theorem 5.2. With the strict failure function, at most $1 + \log_\varphi m$ comparisons are performed against any single text character, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

Proof. Consider the comparisons against a text character $T[i]$, starting from the current value $q_0 \leq m-1$ of q; after an occurrence, q is reset to $f(m) < m$ before the next character is read. They follow the strict failure chain $q_0 > q_1 > \cdots > q_t = 0$, where $q_{s+1} = f'(q_s)$, and there are at most $t+1$ of them: one at each q_s, the last of which either succeeds or fails at $q_t = 0$. By Lemma 5.5, $q_s > q_{s+1} + q_{s+2}$ whenever $q_{s+2} \geq 1$, that is, for $s \leq t-3$. Let $F_1 = F_2 = 1$ and $F_k = F_{k-1} + F_{k-2}$ be the Fibonacci numbers. We claim that $q_{t-s} + 1 \geq F_{s+2}$ for $1 \leq s \leq t$. Indeed, $q_{t-1} + 1 \geq 2 = F_3$ and $q_{t-2} + 1 \geq 3 = F_4$ since the chain is strictly decreasing, and for $s \geq 3$,

$$q_{t-s} + 1 \;\geq\; (q_{t-s+1} + 1) + (q_{t-s+2} + 1) \;\geq\; F_{s+1} + F_s \;=\; F_{s+2} \;.$$

Hence $m \geq q_0 + 1 \geq F_{t+2} \geq \varphi^t$, using the bound $F_k \geq \varphi^{k-2}$, which holds for $k = 1, 2$ and propagates because $\varphi^2 = \varphi + 1$. Thus $t \leq \log_\varphi m$, and the number of comparisons is at most $1 + \log_\varphi m$. □

The bound is attained up to an additive constant, on the Fibonacci strings [18]. Let $w_0 = \texttt{b}$, $w_1 = \texttt{a}$, and $w_j = w_{j-1}w_{j-2}$ for $j \geq 2$, so that $w_2 = \texttt{ab}$, $w_3 = \texttt{aba}$, $w_4 = \texttt{abaab}$, $w_5 = \texttt{abaababa}$, and $|w_j| = F_{j+1}$.

Theorem 5.3. Let $P = w_k$ with $k \geq 3$, and let $m = |P|$. In state $q = m-2$, on a text character that is neither `a` nor `b`, KMP performs $k - 1 \geq \log_\varphi m - 1$ comparisons.

Proof. We first record three properties of the strings w_j.

First, for $j \geq 3$, the strings $w_{j-1}w_{j-2}$ and $w_{j-2}w_{j-1}$ differ exactly by the exchange of their last two letters. For $j = 3$ they are `aba` and `aab`; for $j > 3$, both $w_{j-1}w_{j-2} = w_{j-2}\,w_{j-3}\,w_{j-2}$ and $w_{j-2}w_{j-1} = w_{j-2}\,w_{j-2}\,w_{j-3}$ start with w_{j-2}, and what follows differs by the exchange, by induction. In particular w_j ends like w_{j-2}: with `ab` for even j and with `ba` for odd j.

Second, for $j \geq 1$, w_j contains F_j letters `a` and F_{j-1} letters `b`, by induction, and consecutive Fibonacci numbers are coprime, since $\gcd(F_j, F_{j-1}) = \gcd(F_{j-1}, F_{j-2}) = \cdots = \gcd(F_2, F_1) = 1$. Hence w_j is not a power z^r with $r \geq 2$ of a shorter string, as the two letter counts of such a power are both divisible by r.

Third, let g_j be w_j deprived of its last two letters, for $j \geq 2$, so that $|g_j| = F_{j+1} - 2$. For $j \geq 4$, dropping the last two letters of $w_j = w_{j-1}w_{j-2}$ and of the equal-up-to-the-exchange $w_{j-2}w_{j-1}$ gives

$$g_j \;=\; w_{j-1}\, g_{j-2} \;=\; w_{j-2}\, g_{j-1} \;,$$

so g_{j-1} is a suffix of g_j, and also a prefix, being a prefix of w_{j-1}: it is a border of g_j, and $|g_j| - |g_{j-1}| = F_{j-1}$ is a period of g_j. It is the smallest period once $j \geq 5$. Indeed, a period $p < F_{j-1}$ would satisfy $p + F_{j-1} \leq F_{j+1} - 2 = |g_j|$, as $F_{j-2} \geq 2$, so by Lemma 5.4 the number $g = \gcd(p, F_{j-1})$ would be a period of g_j with $g < F_{j-1}$ and $g \mid F_{j-1}$; the prefix w_{j-2} of g_j, of length F_{j-1}, would then be a power of its own prefix of length g, which the second property forbids. Hence $f(|g_j|) = |g_{j-1}|$ in $P = w_k$ for $5 \leq j \leq k$, and directly $f(|g_4|) = f(3) = 1 = |g_3|$ and $f(|g_3|) = f(1) = 0 = |g_2|$.

These borders are strict. The letter following the prefix g_j in P is the first of the last two letters of w_j, which is `a` for even j and `b` for odd j by the first property; so the letters following g_j and g_{j-1} differ, and $f'(|g_j|) = |g_{j-1}|$ for $3 \leq j \leq k$. The strict failure chain at $q = |g_k| = m - 2$ is therefore $|g_{k-1}|, |g_{k-2}|, \ldots, |g_2| = 0$, and a text character outside $\{\mathtt{a}, \mathtt{b}\}$ fails at each of its $k - 1$ states. Finally $m = F_{k+1} \leq \varphi^k$, so $k \geq \log_\varphi m$. □

The third letter is needed. Consecutive states of a strict chain are followed by different pattern letters, so over a binary alphabet a text letter agrees with one of the first two candidates: the delay of KMP on a binary alphabet is at most 2, whatever the pattern.

It remains to compute f'. No new pass over the pattern is needed: f' is obtained from f in a single loop.

Algorithm: Compute Strict Failure Function

1. Compute the failure function f of P.
2. For $q \leftarrow 1$ to $m - 1$:
   (a) If $f(q) = 0$, set $f'(q) \leftarrow 0$.
   (b) Else if $P[f(q) + 1] \neq P[q + 1]$, set $f'(q) \leftarrow f(q)$.
   (c) Else set $f'(q) \leftarrow f'(f(q))$.
3. Set $f'(m) \leftarrow f(m)$.

Lemma 5.6. The algorithm above computes the strict failure function in $O(m)$ time.

Proof. For correctness, we argue by strong induction on q. The candidates for $f'(q)$ are exactly the elements of the failure chain at q, in decreasing order, and we must return the first one k with $P[k + 1] \neq P[q + 1]$, or 0. The first candidate is $k = f(q)$; if it is strict, we are done. Otherwise $P[f(q) + 1] = P[q + 1]$, so a border $P[1..k]$ of $P[1..q]$ with $k < f(q)$ is strict for q if and only if it is strict for the shorter prefix $P[1..f(q)]$: both conditions read $P[k + 1] \neq P[q + 1] = P[f(q) + 1]$. The remaining candidates are the elements of the failure chain at $f(q)$, so the answer is $f'(f(q))$, which is already computed since $f(q) < q$. Each iteration takes constant time, hence $O(m)$ in total on top of the $O(m)$ of Lemma 5.2. □

Both MP and KMP thus run in $O(n + m)$ total time; the difference is the worst-case work per text character, which drops from $\Theta(m)$ to $O(\log m)$. This distinction matters when the text arrives as a stream and each character must be processed before the next one arrives. In the next lecture we push this direction to its natural limit: an automaton that spends *exactly one* transition per text character.

October 5, 2026

Lecture 6: String Algorithms II

The Knuth–Morris–Pratt algorithm reduced the worst-case work per text character from $\Theta(m)$ to $O(\log m)$. This lecture pushes into two different directions. First, we eliminate the failure iterations altogether: the string-matching automaton spends *exactly one* transition per text character, and Simon's algorithm shows how to store this automaton in $O(m)$ space, independently of the alphabet. Second, we give up on reading every text character: the Boyer–Moore algorithm scans each window from right to left and can skip parts of the text without reading them. Throughout, Σ denotes the alphabet and $\sigma = |\Sigma|$ its size; f is the failure function of the previous lecture.

6.1 The String-Matching Automaton

In state q, on reading a character a, the MP search walks down the failure chain of q until it reaches a border that a extends. Where this walk ends depends only on q and a, not on the rest of the text. So we can determine it once for every pair (q, a), store the results in a table, and replace the walk by a single lookup.

Definition 6.1: String-matching automaton

The *string-matching automaton* of $P[1..m]$ is the deterministic finite automaton with states $\{0, 1, \dots, m\}$, initial state 0, accepting state m, and transition function

$$\delta(q, a) = \max\{k \geq 0 : P[1..k] \text{ is a suffix of } P[1..q]\, a\}$$

for $0 \leq q \leq m$ and $a \in \Sigma$.

In words: having matched $P[1..q]$ and then read a, the new state is the length of the longest prefix of P that the text now ends with. For $P = \texttt{ababac}$ over $\Sigma = \{\texttt{a}, \texttt{b}, \texttt{c}\}$, the table is

q	0	1	2	3	4	5	6
$\delta(q, \texttt{a})$	1	1	3	1	5	1	1
$\delta(q, \texttt{b})$	0	2	0	4	0	4	0
$\delta(q, \texttt{c})$	0	0	0	0	0	6	0

For instance, $\delta(5, \texttt{b}) = 4$ because the longest prefix of $\texttt{ababac}$ that is a suffix of $\texttt{ababab}$ is $\texttt{abab}$; the automaton is drawn in Figure 6.1. On the text $T = \texttt{abababac}$ the states are $0, 1, 2, 3, 4, 5, 4, 5, 6$. At $T[6] = \texttt{b}$, where Morris–Pratt first failed from 5 to $f(5) = 3$ and then advanced to 4, the automaton takes the single transition $\delta(5, \texttt{b}) = 4$. The search needs nothing but the table.

Algorithm: Automaton Search

1. Compute the table δ of P.
2. Set $q \leftarrow 0$.
3. For $i \leftarrow 1$ to n: set $q \leftarrow \delta(q, T[i])$; if $q = m$, report an occurrence at position $i - m + 1$.

Theorem 6.1. After reading $T[1..i]$, the state of Automaton Search is the length of the longest prefix of P that is a suffix of $T[1..i]$. Hence the algorithm reports exactly the occurrences of P in T.

Proof. By induction on i; the case $i = 0$ is clear. Let q be the state after $T[1..i-1]$, let $a = T[i]$, and let k be the length of the longest prefix of P that is a suffix of $T[1..i]$. We show $k = \delta(q, a)$.

$\delta(q, a) \le k$. By the induction hypothesis, $P[1..q]$ is a suffix of $T[1..i-1]$, so $P[1..q]\,a$ is a suffix of $T[1..i]$. Every prefix of P that is a suffix of $P[1..q]\,a$ is thus a suffix of $T[1..i]$, and has length at most k.

$k \le \delta(q, a)$. This is clear if $k = 0$. Otherwise $P[k] = a$, and $P[1..k-1]$ is a prefix of P and a suffix of $T[1..i-1]$, so $k - 1 \le q$ by the induction hypothesis. Since $P[1..k-1]$ and $P[1..q]$ are both suffixes of $T[1..i-1]$, the shorter is a suffix of the longer, and therefore $P[1..k]$ is a suffix of $P[1..q]\,a$.

Finally, P occurs at position $i - m + 1$ if and only if P is a suffix of $T[1..i]$, that is, if and only if the state after $T[1..i]$ is m. □

It remains to compute the table. The failure function does it, by the very walk the table is meant to replace.

Lemma 6.1. For every state q and every character a:

1. if $q < m$ and $a = P[q+1]$, then $\delta(q, a) = q + 1$;
2. otherwise, if $q \ge 1$, then $\delta(q, a) = \delta(f(q), a)$;
3. otherwise, that is, if $q = 0$ and $a \ne P[1]$, then $\delta(q, a) = 0$.

Proof. In case 1, $P[1..q+1]$ is a suffix of $P[1..q]\,a$, and no longer prefix of P can be, so $\delta(q, a) = q + 1$. Case 3 is immediate: the only nonempty suffix of the one-letter string a is a, which is not $P[1]$.

In case 2, let $P[1..k]$ with $k \ge 1$ be a suffix of $P[1..q]\,a$. If $k = q + 1$, then $q < m$ and $a = P[q+1]$, which is case 1; hence $k \le q$. Then $P[1..k-1]$ is a suffix of $P[1..q]$ of length less than q, that is, a border of $P[1..q]$, and every border is a suffix of the longest border $P[1..f(q)]$. Thus $P[1..k]$ is a suffix of $P[1..f(q)]\,a$. Conversely, every prefix of P that is a suffix of $P[1..f(q)]\,a$ is a suffix of $P[1..q]\,a$, because $P[1..f(q)]$ is a suffix of $P[1..q]$. The two maxima agree: $\delta(q, a) = \delta(f(q), a)$. □

Rule 2 is one step of MP's failure walk: the comparison with $P[q+1]$ fails, or there is no $P[m+1]$ to compare with, and the search continues from the border $f(q)$. Since $f(q) < q$, the entry it refers to is already known when the states are processed in increasing order.

Algorithm: Build Transition Table

1. Compute the failure function f of P.
2. For $q \leftarrow 0$ to m, for each $a \in \Sigma$:
   (a) If $q < m$ and $a = P[q+1]$, set $\delta(q, a) \leftarrow q + 1$.
   (b) Else if $q \ge 1$, set $\delta(q, a) \leftarrow \delta(f(q), a)$.
   (c) Else set $\delta(q, a) \leftarrow 0$.

The table has $(m+1)\sigma$ entries, each filled in constant time, so the preprocessing takes $O(m\sigma)$ time and space, and the search takes $O(n)$ time. More precisely, the search has *delay exactly* 1: one table lookup per text character, with no amortization needed. This is perfectly fine for small alphabets. For large alphabets (Unicode, say, or matching over machine words), the factor σ in the preprocessing is the problem that Simon's algorithm removes.

6.2 Simon's Algorithm

Almost all of the transition table is redundant. Call the transitions $(q, P[q+1]) \mapsto q+1$ for $0 \le q < m$ *forward*; they trace the spine of the automaton. Call a transition $(q, a) \mapsto p$ *significant* if it is not forward and $p \ge 1$. All remaining transitions lead to state 0, and there is no need to store them: if a lookup finds neither a forward nor a significant transition, the automaton moves to state 0.

Note that a significant transition satisfies $p \le q$: if $a = P[q+1]$ and $q < m$, the transition is forward; otherwise, by Lemma 6.1, $p \le f(q) + 1 \le q$.

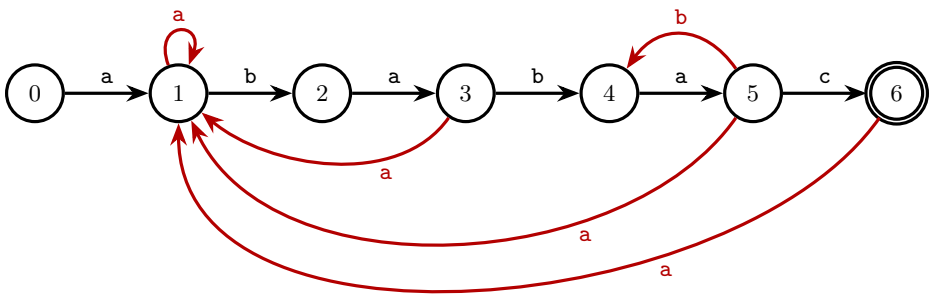


Figure 6.1: The string-matching automaton of $P = \texttt{ababac}$ over $\Sigma = \{\texttt{a}, \texttt{b}, \texttt{c}\}$. Forward transitions in black, significant transitions in red; all missing transitions lead to state 0 and are not drawn. There are $m = 6$ forward and $5 \le m$ significant transitions.

There are exactly m forward transitions. The surprise, due to Simon [20], is that the significant transitions are equally rare, no matter the alphabet.

Theorem 6.2. The string-matching automaton of a pattern of length m has at most m significant transitions.

Proof. Let $(q, a) \mapsto p$ be a significant transition. Since $p \ge 1$, the definition of δ gives that $P[1..p]$ is a suffix of $P[1..q]\,a$. Hence $P[p] = a$, and $P[1..p-1]$ is a suffix of $P[1..q]$, that is, $P[j] = P[j+s]$ for $1 \le j \le p-1$ with $s = q - p + 1$. Substituting $i = j + s$, which ranges from $s+1$ to $s+p-1 = q$, this reads

$$P[i] = P[i-s] \quad \text{for all } i \in \{s+1, \ldots, q\}. \tag{6}$$

Since $p \le q$, the *shift* s lies in $\{1, \ldots, m\}$. We show that the map $(q, a) \mapsto s$ is injective on significant transitions; the bound follows.

Suppose two significant transitions $(q, a) \mapsto p$ and $(q', a') \mapsto p'$ have the same shift s, that is, $q - p + 1 = s = q' - p' + 1$. If $q = q'$, then $p = q - s + 1 = q' - s + 1 = p'$ and $a = P[p] = P[p'] = a'$, so the transitions coincide. Now assume $q > q'$, so in particular $q' < m$. Because the transition from q' is significant and $q' < m$, we must have $a' \ne P[q'+1]$: otherwise the transition on a' would be the forward one. On the other hand, $p' = q' - s + 1 \ge 1$ gives $q' \ge s$, so $i = q' + 1$ lies in $\{s+1, \ldots, q\}$ and (6) for the *first* transition yields

$$P[q'+1] = P[q'+1-s] = P[p'] = a' \ ,$$

a contradiction. □

Simon's algorithm stores, for each state q, the list $\mathrm{sig}(q)$ of its significant transitions, as pairs (a, r) with $\delta(q, a) = r$, ordered by decreasing target r; by Theorem 6.2 this takes $O(m)$ space in total. Two pairs of the same list have different targets: $(a, r) \in \mathrm{sig}(q)$ implies $P[r] = a$, and their characters differ.

Algorithm: Simon Search

1. Compute the lists $\mathrm{sig}(0), \ldots, \mathrm{sig}(m)$ of P.
2. Set $q \leftarrow 0$.
3. For $i \leftarrow 1$ to n: let $a \leftarrow T[i]$.
   (a) If $q < m$ and $a = P[q+1]$, set $q \leftarrow q + 1$.
   (b) Otherwise, scan $\mathrm{sig}(q)$ in order, comparing a with the character of each pair; at the first pair (a, r), set $q \leftarrow r$.
   (c) If no pair of $\mathrm{sig}(q)$ has the character a, set $q \leftarrow 0$.

   If $q = m$, report an occurrence at position $i - m + 1$.

The three cases are the forward transition, a significant one, and a transition to 0, so each iteration performs $q \leftarrow \delta(q, T[i])$: this is Automaton Search with a compressed table, and it is correct by Theorem 6.1. The order of the lists plays no role for correctness. It matters for the running time: a single iteration may scan a whole list, but with this order all scans together cost $O(n)$, as Theorem 6.3 below shows.

The lists are built from the failure function, without ever materializing the full table. By Lemma 6.1, for $a \neq P[q+1]$ the transition of q on a is the transition of $f(q)$ on a, which is the forward transition of $f(q)$, or a significant transition of $f(q)$, or leads to 0. Hence, for $1 \leq q \leq m$,

$$\mathrm{sig}(q) = \big\{(a, r) \in \mathrm{sig}(f(q)) \cup \{(P[f(q)+1],\ f(q)+1)\} : a \neq P[q+1]\big\}\ , \tag{7}$$

where the condition $a \neq P[q+1]$ is void for $q = m$, and $\mathrm{sig}(0) = \emptyset$. For $P = \texttt{ababac}$ and $q = 5$, we have $f(5) = 3$ and $\mathrm{sig}(3) = \{(\texttt{a}, 1)\}$; the forward transition of state 3 contributes $(\texttt{b}, 4)$, and nothing is removed because neither pair has the character $P[6] = \texttt{c}$. Hence $\mathrm{sig}(5)$ consists of $(\texttt{b}, 4)$ followed by $(\texttt{a}, 1)$, as in Figure 6.1.

Algorithm: Build Significant Transitions

1. Compute the failure function f of P, and set $\mathrm{sig}(0) \leftarrow \emptyset$.
2. For $q \leftarrow 1$ to m:
   (a) Set $\mathrm{sig}(q)$ to the pair $(P[f(q)+1],\ f(q)+1)$ followed by a copy of $\mathrm{sig}(f(q))$.
   (b) If $q < m$, scan $\mathrm{sig}(q)$ and delete the pair with character $P[q+1]$, if there is one.

Lemma 6.2. The algorithm above computes all lists of significant transitions, each ordered by decreasing target, in $O(m)$ time.

Proof. Step 2 implements (7), and $\mathrm{sig}(f(q))$ is available when it is needed because $f(q) < q$. The targets in $\mathrm{sig}(f(q))$ are at most $f(q)$, so the pair $(P[f(q)+1],\ f(q)+1)$ has the largest target, and the order is maintained. The copy and the scan of iteration q take $O(1 + |\mathrm{sig}(f(q))|)$ time. The iteration adds one pair and deletes at most one, because the characters in the list are pairwise distinct. Hence

$|\mathrm{sig}(q)| \geq |\mathrm{sig}(f(q))|$, and therefore

$$\sum_{q=1}^{m} |\mathrm{sig}(f(q))| \;\leq\; \sum_{q=1}^{m} |\mathrm{sig}(q)| \;\leq\; m$$

by Theorem 6.2. □

Theorem 6.3. On a text of length $n \geq 1$, step 3 of Simon Search performs at most $2n - 1$ character comparisons. Hence Simon's algorithm finds all occurrences of P in T in $O(m + n)$ time and $O(m)$ space, for every alphabet.

Proof. Call a comparison *successful* if the two characters are equal, and *unsuccessful* otherwise. An iteration stops scanning at its first successful comparison, so there are at most n successful comparisons.

For the unsuccessful ones, let q_i be the state before iteration i, and let $g_i = i - q_i$ be the text position at which the current partial match $T[g_i..i-1] = P[1..q_i]$ starts; thus $g_1 = 1$. Every comparison of iteration i tests one candidate r for the next state: $r = q_i + 1$ in step (a), and the target r of the pair in step (b). Charge an unsuccessful comparison to the text position $i + 1 - r$, at which a match of length r ending at i would start.

The candidates are tested in order of decreasing r, since the targets in $\mathrm{sig}(q_i)$ are at most q_i and the list is ordered by decreasing target. The next state q_{i+1} is smaller than every rejected candidate: it is either the candidate of the successful comparison, or 0. So the positions charged in iteration i are pairwise distinct, and from $q_{i+1} < r \leq q_i + 1$ each of them satisfies

$$g_i \;\leq\; i + 1 - r \;<\; i + 1 - q_{i+1} = g_{i+1} \;\;.$$

The ranges $\{g_i, \ldots, g_{i+1} - 1\}$ of different iterations are disjoint, so every text position is charged at most once, and all charged positions are smaller than $g_{n+1} = n + 1 - q_{n+1}$. Hence there are at most $n - q_{n+1}$ unsuccessful comparisons.

If the last iteration has a successful comparison, then $q_{n+1} \geq 1$ and there are at most $n - 1$ unsuccessful comparisons; otherwise there are at most $n - 1$ successful ones. In both cases the total is at most $2n - 1$. An iteration takes constant time plus time proportional to its number of comparisons, so the search takes $O(n)$ time, and the lists are built in $O(m)$ time by Lemma 6.2. □

The bound $2n - 1$ is due to Hancart [21]; it is attained by $P = \mathtt{ab}$ and $T = \mathtt{a}^n$: every iteration after the first compares $T[i] = \mathtt{a}$ with $P[2] = \mathtt{b}$ and then with the pair $(\mathtt{a}, 1)$ of $\mathrm{sig}(1)$. Unlike the full automaton, Simon Search does not have delay 1: an iteration in state q performs up to $1 + |\mathrm{sig}(q)|$ comparisons instead of one lookup. This is at most σ, because the characters of $\mathrm{sig}(q)$ are pairwise distinct and differ from $P[q + 1]$; Hancart also shows that it is at most $1 + \log_2 m$. In particular, Simon Search has constant delay on an alphabet of constant size, which KMP does not (Theorem 5.3).

6.3 Boyer–Moore

All algorithms so far read every text character at least once. To perform fewer than n comparisons, an algorithm has to *conclude* something about text characters it never reads. The Boyer–Moore (BM) algorithm [22] achieves this with one twist: it compares the pattern with the text *from right to left*.

BM aligns P with a *window* $T[j + 1..j + m]$ of the text and compares from right to left: $P[m]$ with $T[j + m]$, then $P[m - 1]$ with $T[j + m - 1]$, and so on. The window *ends* at the first mismatch, or as an

occurrence when all of P has matched. The pattern is then shifted to the right by some $d \geq 1$, and the scan restarts at the right end $j + d + m$ of the new window. Two rules determine d; Figure 6.2 shows both.

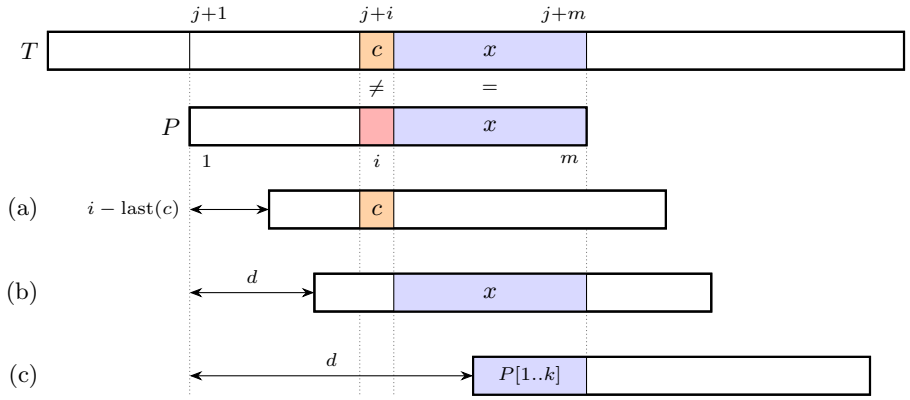


Figure 6.2: A Boyer–Moore window at text positions $j+1, \ldots, j+m$ after a mismatch at pattern position i: the suffix $x = P[i+1..m]$ has been matched, and $c = T[j+i] \neq P[i]$. (a) The bad-character rule places the last occurrence of c in P below $T[j+i]$; drawn for $\text{last}(c) < i$. (b), (c) The good-suffix rule takes the smallest shift d for which the shifted pattern agrees with x where the two overlap: either x occurs again in P, or a prefix $P[1..k]$ with $k = m - d$ is a suffix of x.

The bad-character rule. For each $a \in \Sigma$, let

$$\text{last}(a) = \max\big(\{i : P[i] = a\} \cup \{0\}\big) \ .$$

Suppose the comparison fails at pattern position i, reading the text character $c = T[j+i] \neq P[i]$. In any alignment that survives, the text character c must face a pattern position holding the character c (Figure 6.2(a)). The closest opportunity to the left of position i is $\text{last}(c)$ restricted to positions before i; the simplified rule uses $\text{last}(c)$ directly and shifts by

$$\max\{1,\ i - \text{last}(c)\} \ .$$

If c does not occur in P at all, the window jumps clear over position $j + i$: a shift by i, which equals m when the very first comparison fails. If no character of T occurs in P, every window costs one comparison and shifts by m, and the search performs $\lfloor n/m \rfloor$ comparisons.

The good-suffix rule. The bad-character rule alone gives no worst-case guarantee. For $P = \texttt{ba}^{m-1}$ and $T = \texttt{a}^n$, every window matches $m - 1$ characters and fails at $i = 1$ with $c = \texttt{a}$; since $\text{last}(\texttt{a}) = m$, the shift is 1, and the search performs $(n - m + 1)\, m$ comparisons. The second rule uses the structure of the pattern instead. Suppose the comparison fails at position i, after the suffix $x = P[i+1..m]$ has been matched; x is empty if $i = m$, and we also allow $i = 0$ for a full match. A shift by d is consistent with what has been read only if the pattern, moved d places to the right, still agrees with x wherever the two overlap:

$$P[j-d] = P[j] \qquad \text{for all } j \in \{i+1, \ldots, m\} \text{ with } j - d \geq 1 \ .$$

The *good-suffix shift* $\text{gs}(i)$ is the smallest $d \geq 1$ with this property; $d = m$ always qualifies. For $P = \texttt{ba}^{m-1}$ and $i = 1$, no $d < m$ qualifies, because $P[1+d] = \texttt{a} \neq \texttt{b} = P[1]$; hence $\text{gs}(1) = m$, and the windows of the

example above no longer overlap. In terms of $k = m - d$, the property says that x is a suffix of $P[1..k]$ if $k \geq |x|$ (Figure 6.2(b)), and that $P[1..k]$ is a suffix of x otherwise (Figure 6.2(c)), so

$$\mathrm{gs}(i) = m - \max\bigl\{k \in \{0, \ldots, m-1\} : x \text{ is a suffix of } P[1..k] \text{ or } P[1..k] \text{ is a suffix of } x\bigr\} \ . \tag{8}$$

The table is computed in $O(m)$ time from two failure functions: the failure function f of P, and the failure function f^R of the reversed pattern $P^R = P[m]\,P[m-1]\cdots P[1]$.

Lemma 6.3. For $0 \leq i \leq m$,

$$\mathrm{gs}(i) = \min\Bigl(\{m - f(m)\} \cup \bigl\{\ell - f^R(\ell) : 1 \leq \ell \leq m \text{ and } f^R(\ell) = m - i\bigr\}\Bigr) \ .$$

Proof. Let $b = |x| = m - i$. Every candidate k in (8) yields the shift $m - k$, and there are two kinds of candidates.

$P[1..k]$ is a suffix of x. Then $P[1..k]$ is a border of P, so $k \leq f(m)$. The value $k = f(m)$ is itself a candidate: $P[1..f(m)]$ and x are both suffixes of P, so the shorter one is a suffix of the longer one. Hence no candidate of this kind beats the shift $m - f(m)$, and this shift is available.

x is a suffix of $P[1..k]$. Then x occurs in P ending at position $k < m$. Read from right to left, and since $x^R = P^R[1..b]$, this says that $P^R[1..b]$ is a suffix of $P^R[1..\ell]$ for $\ell = m - k + b$, with $b < \ell \leq m$. So b is the length of a border of $P^R[1..\ell]$, and the shift is $m - k = \ell - b$. Conversely, every such ℓ comes from the candidate $k = m - \ell + b$.

It remains to show that the smallest value of $\ell - b$ is attained with $b = f^R(\ell)$. If b is the length of a border of $P^R[1..\ell]$ other than the longest one, let ℓ' be the element just above b in the failure chain of ℓ. Then $f^R(\ell') = b$ and $\ell' < \ell$, so $\ell' - b < \ell - b$. Hence the best shift of the second kind is the minimum of $\ell - f^R(\ell)$ over all ℓ with $f^R(\ell) = b = m - i$. □

Algorithm: Compute Good-Suffix Table

1. Compute the failure function f of P and the failure function f^R of P^R.
2. For $i \leftarrow 0$ to m: set $\mathrm{gs}(i) \leftarrow m - f(m)$.
3. For $\ell \leftarrow 1$ to m: let $b \leftarrow f^R(\ell)$ and set $\mathrm{gs}(m - b) \leftarrow \min\bigl(\mathrm{gs}(m - b),\ \ell - b\bigr)$.

The table is correct by Lemma 6.3 and takes $O(m)$ time by Lemma 5.2. The search combines the two rules by always taking the larger shift.

Algorithm: Boyer–Moore Search

1. Compute $\mathrm{last}(a)$ for all $a \in \Sigma$ and $\mathrm{gs}(i)$ for $0 \leq i \leq m$.
2. Set $j \leftarrow 0$.
3. While $j + m \leq n$:
   (a) Set $i \leftarrow m$; while $i \geq 1$ and $P[i] = T[j + i]$, decrement i.
   (b) If $i = 0$, report an occurrence at position $j + 1$ and set $j \leftarrow j + \mathrm{gs}(0)$.
   (c) Otherwise, set $j \leftarrow j + \max\bigl(\mathrm{gs}(i),\ i - \mathrm{last}(T[j + i])\bigr)$.

Example. Let $P = \texttt{aabaa}$, so that $\text{last}(\texttt{a}) = 5$, $\text{last}(\texttt{b}) = 3$, and $\text{last}(c) = 0$ for every other character c. The failure function is $f = (0, 1, 0, 1, 2)$, so the default shift is $m - f(m) = 3$. The reversed pattern is P itself, so $f^R = f$, and the loop over ℓ lowers $\text{gs}(5)$ to $1 - 0 = 1$ at $\ell = 1$ and $\text{gs}(4)$ to $2 - 1 = 1$ at $\ell = 2$, while the other entries keep the default:

i	0	1	2	3	4	5
$\text{gs}(i)$	3	3	3	3	1	1

For instance, $\text{gs}(4) = 1$ because after matching the final $\texttt{a}$, a shift by 1 places $P[4] = \texttt{a}$ below it, and $\text{gs}(3) = 3$ because after matching $\texttt{aa}$, shifts by 1 and 2 would place $\texttt{ba}$ and $\texttt{ab}$ below it, whereas a shift by 3 places the prefix $\texttt{aa}$ there. The search in $T = \texttt{abaabaacaabaab}$ inspects four windows:

window	comparisons	outcome	candidate shifts	shift
$T[1..5] = \texttt{abaab}$	1	$i = 5$, $T[5] = \texttt{b}$	$\text{gs}(5) = 1$, $5 - \text{last}(\texttt{b}) = 2$	2
$T[3..7] = \texttt{aabaa}$	5	occurrence at 3	$\text{gs}(0) = 3$	3
$T[6..10] = \texttt{aacaa}$	3	$i = 3$, $T[8] = \texttt{c}$	$\text{gs}(3) = 3$, $3 - \text{last}(\texttt{c}) = 3$	3
$T[9..13] = \texttt{aabaa}$	5	occurrence at 9	$\text{gs}(0) = 3$	3

The text has 14 characters and the search performs 14 comparisons; $T[1]$, $T[2]$, and $T[14]$ are never read.

The rule defined above is the *weak* good-suffix rule. The *strong* rule also uses the mismatch itself: the text character below position i is not $P[i]$, so a shift by d with $i - d \geq 1$ is useless unless $P[i - d] \neq P[i]$. It is computed along the same lines, and it is the version to which the following worst-case bound of Cole [24] applies.

Theorem 6.4. With the strong good-suffix rule, if P does not occur in T, then the Boyer–Moore search performs at most $3n$ character comparisons. There are inputs on which it performs $3n - o(n)$.

We do not prove Theorem 6.4; the proof is long, and the bound was only settled two decades after the algorithm was published (Knuth, Morris, and Pratt [18] had shown $6n$). The assumption that P does not occur cannot be dropped. For $P = \texttt{a}^m$ and $T = \texttt{a}^n$, every window is an occurrence, the shift after it is $\text{gs}(0) = \text{per}(P) = 1$, and the next window re-reads $m - 1$ characters that are already known to match, for $\Theta(nm)$ comparisons in total. The remedy, due to Galil [23], is the memory we will meet again in the two-way algorithm: after an occurrence, the shift by $\text{per}(P)$ leaves the first $m - \text{per}(P)$ characters of the new window known to match, and the right-to-left scan stops when it reaches them. With this change the search performs $O(n)$ comparisons on every input.

A variant with a simple bound. Boyer–Moore is hard to analyze because a window forgets what the previous windows matched and may compare the same text characters again. Apostolico and Giancarlo [25] proposed a variant that remembers, and for which a linear bound has a short proof; we follow the presentation of Gusfield [26].

Two tables are used. The first comes from the pattern: for $1 \leq i \leq m$, let $\text{suff}(i)$ be the length of the longest suffix of $P[1..i]$ that is also a suffix of P, so that $\text{suff}(m) = m$. Comparing $P[1..i]$ with P from the right for each i computes the table in $O(m^2)$ time. It can be computed in $O(m)$ time [26, 27], by an algorithm that we do not present. The second table is filled in during the search: $M(r)$, for a text position r,

records that a suffix of P of length $M(r)$ ends at r, that is, $T[r-M(r)+1..r] = P[m-M(r)+1..m]$. Initially $M(r) = 0$ for all r; a value is set only at the right end of each window, at the end of that window. If $M(r) > 0$, we say that r *covers* the interval $[r - M(r) + 1, r]$.

The search is Boyer–Moore Search, except that at a text position with $M(t) > 0$ it skips instead of comparing. Each iteration of step 2 treats one window $T[j + 1..j + m]$; the window ends when step (b) does.

Algorithm: Apostolico–Giancarlo Search

1. Compute last, gs and suff; set $M(r) \leftarrow 0$ for $1 \le r \le n$, and $j \leftarrow 0$.
2. While $j + m \le n$:
   (a) Set $i \leftarrow m$.
   (b) Repeat, until $i = 0$ or the loop is stopped: set $t \leftarrow j + i$.
   - If $M(t) = 0$ and $P[i] = T[t]$, set $i \leftarrow i - 1$.
   - If $M(t) = 0$ and $P[i] \neq T[t]$, stop.
   - If $M(t) > 0$, set $s \leftarrow \text{suff}(i)$ and $i \leftarrow i - \min(M(t), s)$, and stop if $s < M(t)$.

   (c) Set $M(j + m) \leftarrow j + m - t$.
   (d) If $i = 0$, report an occurrence at position $j + 1$ and set $j \leftarrow j + \text{gs}(0)$.
   (e) Otherwise, set $j \leftarrow j + \max\big(\text{gs}(i),\ i - \text{last}(T[j + i])\big)$.

Steps (a), (d) and (e) are those of Boyer–Moore Search, and the first two cases of step (b) are its comparisons. The third case skips $\min(M(t), s)$ positions without comparing. With i denoting the value before the skip, both $T[t-M(t)+1..t]$ and $P[i-s+1..i]$ are suffixes of P, so they agree on their last $\min(M(t), s)$ characters: the pattern matches the text on the skipped positions. If $s < M(t)$, the pair below them, if there is one, is known to differ: $T[t-s] = P[m-s]$ lies in the interval covered by t, while $P[i-s] \neq P[m-s]$ by the maximality of s. So step (b) ends with $i = 0$ if the window is an occurrence, and with a mismatch at i otherwise: the algorithm visits the same windows as Boyer–Moore and makes the same shifts.

In step (c), t is the last text position examined in the window, by a comparison or by a lookup of M. The value $M(j + m)$ is valid, since all window positions to the right of t were matched. It can be smaller than the number of matched characters: if the window ends at a position t with $M(t) > 0$, the new interval $[t + 1, j + m]$ excludes the characters skipped at t, although they matched too. The analysis relies on this.

Call a text position *hidden* if it lies in a covered interval without being the right end of that interval. Both notions refer to the current table M. An entry of M is set once and never changed, so intervals are only added, and a hidden position stays hidden.

Lemma 6.4. In Apostolico–Giancarlo Search:

1. after each execution of step (c), any two covered intervals are disjoint or nested;
2. each time step (b) sets $t \leftarrow j + i$, the position t is not hidden.

Proof. The table M changes only in step (c). Consider one iteration of step 2, with $r = j + m$, and suppose that any two covered intervals are disjoint or nested when it starts; this holds for the first iteration, when there is no covered interval. We show claim 2 for this iteration, and claim 1 after its step (c). During step (b), the covered intervals are those created by earlier iterations, and their right ends are smaller than r.

Claim 2. The first position examined is r, which lies in no interval. Now let t be an examined position that is not hidden, and suppose that the loop continues. If $M(t) = 0$, then t lies in no interval, as it is neither hidden nor a right end. The next position is $t - 1$, and an interval that contains $t - 1$

but not t has $t-1$ as its right end; so $t-1$ is not hidden. If $M(t) > 0$, the loop continues only if $s \geq M(t)$, and the next position is $t' = t - M(t)$, just to the left of the interval I covered by t. Suppose t' is hidden in an interval I'. Then I' contains t' and $t'+1$, the left end of I. So I' meets I and is not contained in it, hence $I \subseteq I'$. Thus $t \in I'$, and t is not the right end of I', because the interval with right end t is I. So t is hidden, a contradiction.

Claim 1. Let t be the last position examined in step (b); if $t < r$, step (c) creates the interval $[t+1, r]$. An existing interval that meets $[t+1, r]$ without lying inside it contains t and $t+1$, so t would be hidden, contradicting claim 2. Hence every existing interval that meets $[t+1, r]$ lies inside it, and any two covered intervals are disjoint or nested after step (c). □

Theorem 6.5. Apostolico–Giancarlo Search performs at most $2n$ character comparisons.

Proof. An unsuccessful comparison stops the loop of step (b), so there is at most one per iteration of step 2, and there are at most n iterations, since each increases j by at least 1.

It remains to show that no text position is compared successfully twice. Consider a successful comparison at position $t = j+i$, in the iteration with $r = j+m$. Later in this iteration, only positions smaller than t are examined. If $i = 1$, then $t = j+1$, and every later iteration examines only positions larger than t, because j has increased. If $i \geq 2$, the loop goes on to examine a position smaller than t, so step (c) creates an interval $[t^* + 1, r]$ with $t^* < t$, which contains t. If $t < r$, then t is hidden from then on, and by claim 2 of Lemma 6.4 it is never examined again. If $t = r$, then $M(t) > 0$ from then on, and comparisons are made only where $M = 0$. So at most n comparisons are successful. □

The lookups of M are linear as well: at most one per window ends the window, and after a lookup at t that does not end the window, t is hidden in the interval created at the end of the window, so it is never examined again. As written, the table M has n entries, but only m of them are in use at any time: the iteration for the window $T[j+1..j+m]$ reads $M(t)$ only for $j < t \leq j+m$, and writes only $M(j+m)$. So M is stored in an array of size m, with $M(t)$ in cell $t \bmod m$. When the window moves from j to j', the cells of the positions $t > j+m$ that enter the window are reset to 0, since they still hold the values of earlier positions. The reset takes time proportional to the shift, hence $O(n)$ in total, and the extra space of the algorithm is $O(m+\sigma)$.

GNU `grep` uses a variant of Boyer–Moore.

Where we stand. The algorithms of the last two lectures compare as follows.

algorithm	extra space	search time	delay
Morris–Pratt	$O(m)$	$O(n)$	$\Theta(m)$
Knuth–Morris–Pratt	$O(m)$	$O(n)$	$O(\log m)$
full automaton	$O(m\sigma)$	$O(n)$	1
Simon	$O(m)$	$O(n)$	$\min(\sigma, 1+\log_2 m)$
Boyer–Moore (Galil)	$O(m+\sigma)$	$O(n)$, at best $\lfloor n/m \rfloor$ comparisons	—
Apostolico–Giancarlo	$O(m+\sigma)$	$\leq 2n$ comparisons	—

Every row uses $\Omega(m)$ extra space. The two-way algorithm of Crochemore and Perrin, in the next lecture, runs in $O(n+m)$ time with $O(1)$ extra space.

October 19, 2026

Lecture 7: String Algorithms III

7.1 Two-Way String Matching

KMP keeps an array of m integers, one per pattern position. Can we reduce the space to a handful of integers, independent of the pattern length? The failure function looks essential to KMP, so it is not obvious that we can. The algorithm of Crochemore and Perrin [28] achieves this, and its approach is very different from KMP. It is also the basis of the `strstr` implementation in the GNU C library (glibc) and of substring search in CPython.

Here is the idea. We pick a position ℓ in the pattern and split it in two:

$$P = u \cdot v = P[1..\ell]\ P[\ell{+}1..m].$$

At each alignment of P against T, we scan the right piece v from left to right. Only if v matches in full do we go back and verify the left piece u from right to left (see Figure 7.1). Reading u backward throws away the forward-scanning invariant that KMP maintains, so why do it? Because the split buys something KMP cannot give: when the right phase mismatches, we can often jump forward by many characters at once rather than by a single position.

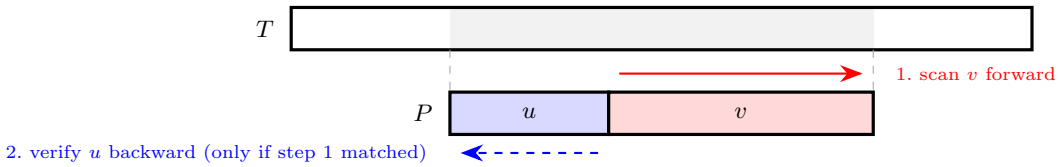


Figure 7.1: One iteration of the two-way algorithm at a fixed alignment. The right phase scans v against the text from left to right; only if v matches in full does the algorithm verify u from right to left.

Let us see what that leap looks like. Take $P =$ `example`, cut at $\ell = 3$, so $u =$ `exa` and $v =$ `mple`, and imagine the right phase matches `m`, `p`, `l` and then fails on the next character. We would like to jump forward by 4, past the three intermediate alignments at offsets 1, 2, 3, and restart scanning at the first character of v. That jump is safe only if none of the three skipped alignments could be an occurrence. Each of them would demand that some suffix of u equal a prefix of v of the same length:

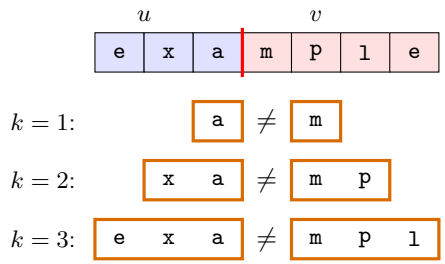


Figure 7.2: The staircase at the cut of $P =$ `example` into $u =$ `exa` and $v =$ `mple`. For each $k \in \{1, 2, 3\}$, inequality (9) compares the last k characters of u (left orange box) with the first k characters of v (right orange box). All three inequalities hold at this cut, so after matching `mpl` the algorithm can safely jump by 4.

None of those three equalities holds at our cut, so we are free to jump by 4 (Figure 7.2). The same reasoning works in general: after matching j characters of v and mismatching on the next, we can safely jump by $j+1$ whenever the pattern satisfies the chain of inequalities

$$P[\ell{-}k{+}1..\ell] \neq P[\ell{+}1..\ell{+}k], \qquad 1 \le k \le j, \tag{9}$$

at the cut. We call this chain the *staircase* at ℓ. The longer it holds, the bigger the jumps we get, so our task is now clear: pick ℓ so the staircase reaches as far as possible.

How far can it reach? Not arbitrarily far. The moment k hits a period of P, the staircase has no choice but to break: if p is a period, then $P[\ell{-}p{+}1..\ell]$ and $P[\ell{+}1..\ell{+}p]$ are two windows of P separated by exactly p positions, and periodicity makes them equal. So the best we can hope for is a cut where the staircase holds for every $k < p$, breaking only at the unavoidable spot $k = p$. Such a cut always exists, whatever the pattern. This is the Critical Factorization Theorem; before stating it precisely we make the words *period* and *critical* formal.

Recall from Lecture 5 that an integer $p \ge 1$ is a *period* of $S[1..k]$ if $S[i] = S[i{+}p]$ whenever $1 \le i \le k{-}p$, that every $p \ge k$ is a period, and that the smallest period is denoted $\mathrm{per}(S)$. For instance, $\mathrm{per}(\texttt{abcabc}) = \mathrm{per}(\texttt{abcab}) = 3$, and $\mathrm{per}(\texttt{abcd}) = 4$ (only the trivial period). Equivalently, p is a period of S if and only if S is a prefix of r^∞ for some string r of length p, and if and only if $S[1..k-p]$ is a border of S.

The staircase lives at a particular cut, so we need a version of periodicity tailored to a cut. Given a cut $P = u \cdot v$ with $\ell = |u|$, a positive integer r is a *local period* of the cut if $P[i] = P[i+r]$ for every index i with $\ell - r + 1 \le i \le \ell$ for which both sides are defined. The smallest such r is the *local period* $r(u, v)$.

That is, $r(u, v)$ is the smallest k at which the staircase (9) breaks, treating positions outside $[1..m]$ as wildcards. Any global period of P is automatically a local period at every cut, so $r(u, v) \le \mathrm{per}(P)$ always. A factorization is *critical* when equality holds: the staircase breaks exactly at $k = \mathrm{per}(P)$, as late as any period can force.

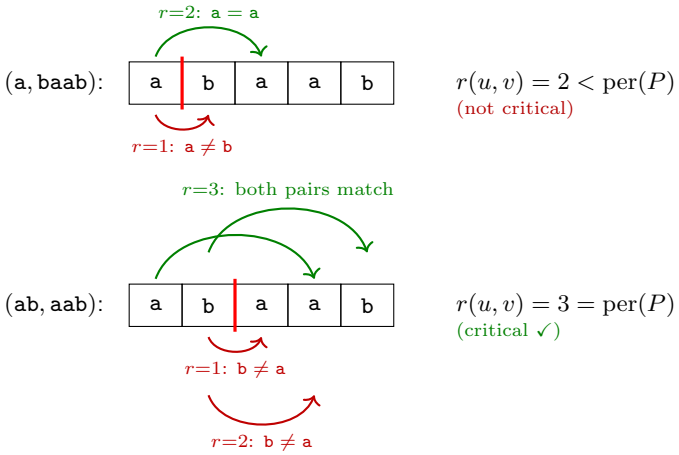


Figure 7.3: The two factorizations of $P = \texttt{abaab}$ (with $\mathrm{per}(P) = 3$). For each cut, red arcs mark the failed local-period tests and the top green arc marks the smallest r that works. Only $(\texttt{ab}, \texttt{aab})$ pushes the local period up to $\mathrm{per}(P)$, so only it is critical.

Example. Consider $P = \texttt{abaab}$, with $\mathrm{per}(P) = 3$; Figure 7.3 compares its two relevant factorizations. At the cut $(\texttt{a}, \texttt{baab})$, the only constraint for $r = 2$ is $P[1] = P[3]$ (the position $i = 0$ is out of range), and $\texttt{a} = \texttt{a}$, so the local period is already 2. At the cut $(\texttt{ab}, \texttt{aab})$, both $r = 1$ and $r = 2$ fail on the pair $P[2] = \texttt{b}$ (which collides with a neighbouring $\texttt{a}$), leaving $r = 3 = \mathrm{per}(P)$: the cut is critical.

Theorem 7.1 (Critical Factorization Theorem). Every nonempty string P has a critical factorization $P = u \cdot v$ with $|u| < \mathrm{per}(P)$.

The proof is constructive and rests on a second kind of order. Fix a total order $\leq$ on the alphabet and let $\preceq$ be the induced lexicographic order on strings: $x \prec y$ if x is a proper prefix of y, or if $x = zax'$ and $y = zby'$ for letters $a < b$. We use two properties of $\preceq$: a proper prefix is strictly smaller, and a common prefix can be cancelled, $zx \preceq zy$ if and only if $x \preceq y$. The *maximal suffix* of P with respect to $\leq$ is the $\preceq$-largest of the m nonempty suffixes of P. The reverse order $\leq'$ on the alphabet, $a <' b$ if and only if $b < a$, induces a second lexicographic order $\preceq'$ and a second maximal suffix. Note that $\preceq'$ is not the reverse of $\preceq$: proper prefixes are smaller in both.

Example. For $P = \texttt{abaab}$ and $\texttt{a} < \texttt{b}$, the suffixes in decreasing order are $\texttt{baab} \succ \texttt{b} \succ \texttt{abaab} \succ \texttt{ab} \succ \texttt{aab}$; the maximal suffix is $\texttt{baab}$, with cut $\ell = 1$. For the reverse order $\texttt{b} <' \texttt{a}$ they order as $\texttt{aab} \succ' \texttt{abaab} \succ' \texttt{ab} \succ' \texttt{baab} \succ' \texttt{b}$, and the maximal suffix is $\texttt{aab}$, with cut $\ell' = 2$. The larger cut is the critical one found in Figure 7.3.

Proof. Let $p = \mathrm{per}(P)$, let v and v' be the maximal suffixes of P for $\leq$ and for $\leq'$, and write $P = uv = u'v'$ with $\ell = |u|$ and $\ell' = |u'|$. We show that the factorization with the larger of the two cuts is critical. By the symmetry between the two orders, we may assume $\ell' \leq \ell$, so that v is a suffix of v', and we show that (u, v) is critical.

First, $\ell < p$. Otherwise $\ell + 1 - p \geq 1$, and the suffix $w = P[\ell{+}1{-}p..m]$ satisfies $w[1..|v|] = P[\ell{+}1{-}p..m{-}p] = P[\ell{+}1..m] = v$ by periodicity, so v is a proper prefix of w and $v \prec w$, contradicting the maximality of v. The same argument gives $\ell' < p$.

Let $r = r(u, v)$ be the local period of the cut. Since p is a local period at every cut, $r \leq p$; suppose $r < p$. Let z be the string of length r realizing the local period: z is a suffix of u or u is a suffix of z, and z is a prefix of v or v is a prefix of z. We treat the four combinations in turn.

z is a suffix of u and a prefix of v. Then zv is a suffix of P, so $zv \preceq v$. Writing $v = zv_1$, this reads $zzv_1 \preceq zv_1$, hence $zv_1 \preceq v_1$, that is, $v \preceq v_1$. If v_1 is empty, this is absurd. Otherwise v_1 is a suffix of P, so $v_1 \preceq v$ by maximality, and $v = v_1$, contradicting $|v_1| < |v|$.

z is a suffix of u and v is a proper prefix of z. Then the suffix zv of P has v as a proper prefix, so $v \prec zv$, again contradicting maximality.

u is a suffix of z and v is a prefix of z. Since $r < p \leq m = |u| + |v|$, the prefix v and the suffix u of z overlap in a string t of length $m - r \geq 1$, which is a suffix of v and a prefix of u. Then t is a prefix and a suffix of $P = uv$, a border of length $m - r$, so r is a period of P and $r \geq p$, a contradiction.

u is a suffix of z and z is a prefix of v. This is the only case that needs the second order. Write $v = zv_1$ and $z = yu$, so that $P = uyuv_1$, and let u_2 be the suffix of u with $u = u'u_2$, so that $v' = u_2v = u_2zv_1$. If v_1 is nonempty, then v_1 and u_2v_1 are suffixes of P, being suffixes of uv_1. Maximality of v gives $v_1 \preceq v$, and maximality of v' gives $u_2v_1 \preceq' u_2zv_1$, that is, $v_1 \preceq' v$ after cancelling u_2. Now v_1 is shorter than v. If it were not a prefix of v, the first position at which the two

differ would carry a smaller letter in v_1 for $\le$ and also for $\le'$, which is absurd. Hence v_1 is a prefix of v, trivially so if v_1 is empty, and $v = zv_1$ has period $r = |z|$. Finally, u is a suffix of $z = v[1..r]$, so $P = uv$ has period r as well: for $i \le \ell$, $P[i] = u[i] = z[r-\ell+i] = v[r-\ell+i] = P[i+r]$, and for $i > \ell$, $P[i] = v[i-\ell] = v[i-\ell+r] = P[i+r]$. Thus $r \ge p$, the final contradiction. □

The cut $\ell = 0$ is possible, but only when P is a power of a single letter: otherwise the first letter of P is not the largest letter of P for one of the two orders, and the corresponding maximal suffix is a proper suffix. In that case $\mathrm{per}(P) = 1$ and the empty u is harmless below.

Computing the two maximal suffixes is the whole preprocessing, and it can be done in one pass with a constant number of registers. The following algorithm, due to Crochemore and Perrin [28], scans P once. It maintains a candidate $P[\ell+1..]$ for the maximal suffix of the prefix read so far, the period p of the candidate, and a competing suffix $P[j+1..]$ that agrees with the candidate on its first $k-1$ characters.

Algorithm: Maximal Suffix

1. Set $\ell \leftarrow 0$, $j \leftarrow 1$, $k \leftarrow 1$, $p \leftarrow 1$.
2. While $j + k \le m$: let $a \leftarrow P[\ell + k]$ and $b \leftarrow P[j + k]$.
   (a) If $b < a$: set $j \leftarrow j + k$, $k \leftarrow 1$, $p \leftarrow j - \ell$.
   (b) If $b = a$ and $k < p$: set $k \leftarrow k + 1$.
   (c) If $b = a$ and $k = p$: set $j \leftarrow j + p$, $k \leftarrow 1$.
   (d) If $b > a$: set $\ell \leftarrow j$, $j \leftarrow \ell + 1$, $k \leftarrow 1$, $p \leftarrow 1$.
3. Return (ℓ, p): the maximal suffix is $v = P[\ell+1..m]$, and $p = \mathrm{per}(v)$.

The three outcomes of a comparison are handled as follows. If the competitor's next character is smaller (case a), the competitor loses, and so does every suffix starting between the competitor and the current position; the next competitor starts at $j + k + 1$, and the candidate, which now extends over the whole matched stretch, has period $j - \ell$. If the characters agree (cases b and c), the match is extended by one position, cycling through the candidate with its period. If the competitor's next character is larger (case d), the competitor beats the candidate and becomes the new candidate. The quantity $\ell + 2j + k$ increases in every iteration and is at most $4m$, so the algorithm runs in $O(m)$ time; it uses four registers. We refer to [28] for the verification of the invariant.

Running the algorithm twice, for $\le$ and for $\le'$, and keeping the pair (ℓ, p) with the larger cut yields a critical position ℓ with $\ell < \mathrm{per}(P)$ by the proof of Theorem 7.1, together with the period of $v = P[\ell+1..m]$. The search below needs $\mathrm{per}(P)$, which is not $\mathrm{per}(v)$ in general. The following lemma shows that the two agree exactly when the period of v extends to all of P, which is checked by comparing $P[1..m-p]$ with $P[p+1..m]$, and that otherwise $\mathrm{per}(P)$ is large.

Lemma 7.1. Let $P = u \cdot v$ be a critical factorization with $|u| < \mathrm{per}(P)$. If $\mathrm{per}(P) \le |v|$, then $\mathrm{per}(v) = \mathrm{per}(P)$.

Proof. Let $p = \mathrm{per}(P)$, $q = \mathrm{per}(v)$ and $\ell = |u|$; clearly $q \le p$. Suppose $q < p$ and set $r = p - q \ge 1$. We show that r is a local period of the cut, contradicting $r(u, v) = p$. Let i be such that $\ell - r + 1 \le i \le \ell$, $i \ge 1$ and $i + r \le m$. Since p is a period of P and $i + p \le \ell + p \le m$, we have $P[i] = P[i + p]$. The positions $i + r = i + p - q$ and $i + p$ both lie in v: the first because $i \ge \ell - r + 1$, the second because $i + p \ge \ell + 1$. As v has period q, $P[i + p - q] = P[i + p]$, and therefore $P[i] = P[i + r]$. □

If P has period $p = \mathrm{per}(v)$, then $\mathrm{per}(P) \le p = \mathrm{per}(v) \le \mathrm{per}(P)$, so $p = \mathrm{per}(P)$; we call this the *periodic case*. Otherwise $\mathrm{per}(v) \ne \mathrm{per}(P)$, so Lemma 7.1 gives $\mathrm{per}(P) > |v| = m - \ell$, and since also $\mathrm{per}(P) > \ell$,

$$\mathrm{per}(P) \;\ge\; q := \max(\ell,\, m - \ell) + 1 \;;$$

we call this the *non-periodic case*. The whole preprocessing takes $O(m)$ time and $O(1)$ space.

With a critical cut in hand, the staircase now tells us exactly how far the algorithm can safely jump in each of its two modes.

Lemma 7.2. Let $P = u \cdot v$ be a critical factorization with $\ell = |u| < p = \mathrm{per}(P)$, and fix a text position $\mathrm{pos} \geq 0$.

(i) If $T[\mathrm{pos}+j] = P[j]$ for $\ell+1 \leq j \leq i-1$ and $T[\mathrm{pos}+i] \neq P[i]$ for some i with $\ell < i \leq m$, then P does not occur at any of the positions $\mathrm{pos}+1+d$ with $1 \leq d < i-\ell$.

(ii) If $T[\mathrm{pos}+j] = P[j]$ for $\ell+1 \leq j \leq m$, then P does not occur at any of the positions $\mathrm{pos}+1+d$ with $1 \leq d < p$.

Proof. In case (ii), set $i = m+1$, so that in both cases $T[\mathrm{pos}+j] = P[j]$ for $\ell+1 \leq j \leq i-1$. Suppose that P occurs at $\mathrm{pos}+1+d$, that is, $T[\mathrm{pos}+d+x] = P[x]$ for $1 \leq x \leq m$. For every x with $\max(1, \ell+1-d) \leq x \leq i-1-d$, the position $\mathrm{pos}+d+x$ lies in the matched region, so

$$P[x] = T[\mathrm{pos}+d+x] = P[x+d] \ . \tag{10}$$

First let $d < p$; this covers case (ii) and part of case (i). We claim that d is a local period of the cut, contradicting $r(u,v) = p > d$. The pairs $(x, x+d)$ defining a local period d are those with $\ell - d + 1 \leq x \leq \ell$, $x \geq 1$ and $x + d \leq m$. Every such x satisfies $x \geq \max(1, \ell+1-d)$ and $x \leq \min(\ell, m-d) \leq i-1-d$, since $\ell \leq i-1-d$ in case (i) and $m-d = i-1-d$ in case (ii). So all these pairs are covered by (10), and the claim follows.

Now let $d \geq p$ in case (i). If p divides d, then $P[i-d] = P[i]$ by periodicity, where $i-d \geq \ell+1$; but the occurrence gives $T[\mathrm{pos}+i] = P[i-d]$, contradicting the mismatch at position i. Otherwise let $d' \in \{1, \ldots, p-1\}$ with $d \equiv d' \pmod{p}$. By periodicity of P, $P[x+d] = P[x+d']$ for all $x \geq 1$ with $x + d \leq m$, so (10) gives $P[x] = P[x+d']$ for $\max(1, \ell+1-d) \leq x \leq i-1-d$. This range contains every pair defining a local period d', since these have $x \geq \max(1, \ell+1-d') \geq \max(1, \ell+1-d)$ and $x \leq \ell \leq i-1-d$. Hence d' is a local period of the cut with $d' < p$, a contradiction. □

There is a subtlety hiding in case (ii) of Lemma 7.2. In the periodic case, after the right phase has matched in full we shift by $p = \mathrm{per}(P)$, but by periodicity the first $m-p$ characters at the new alignment already agree with the text, since we have just confirmed them in the alignment we are leaving behind. If we re-read them from scratch on every iteration, a highly periodic pattern like `aaaa...a` blows the total work up to $\Theta(nm)$. The fix costs one extra integer, the *memory* s, recording how many leading characters of P are already known to match at the current alignment. The right phase then starts at position $\max(\ell, s)+1$, skipping everything at or below s; after a periodic shift we set $s \leftarrow m-p$, and after any mismatch-driven shift we reset $s \leftarrow 0$. In the non-periodic case, nothing is known about the new alignment after a shift by q, and no memory is needed: the shift is so large that the pattern is moved past more than half of the previous window.

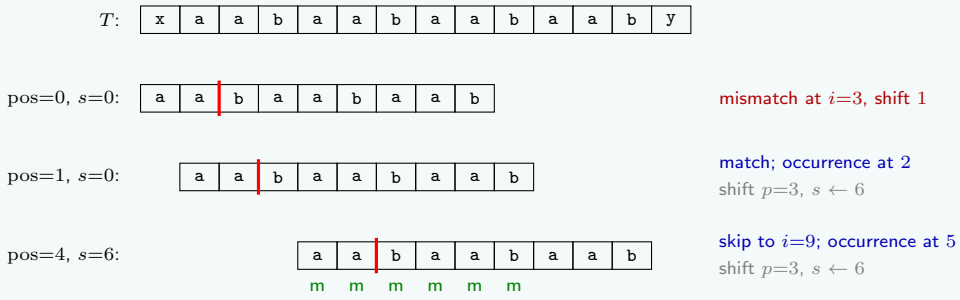
Algorithm: Two-Way Search

1. Preprocess P as above, obtaining the critical position ℓ and, in the periodic case, $p = \mathrm{per}(P)$; in the non-periodic case, set $p \leftarrow \max(\ell, m-\ell)+1$ instead.
2. Set $\mathrm{pos} \leftarrow 0$ and $s \leftarrow 0$.
3. While $\mathrm{pos} + m \leq n$:
   (a) *Right phase.* Set $i \leftarrow \max(\ell, s)+1$; while $i \leq m$ and $P[i] = T[\mathrm{pos}+i]$, increment i. If $i \leq m$, set $\mathrm{pos} \leftarrow \mathrm{pos}+i-\ell$ and $s \leftarrow 0$, and continue with the next iteration.

(b) *Left phase.* Set $j \leftarrow \ell$; while $j > s$ and $P[j] = T[\mathrm{pos} + j]$, decrement j. If $j \le s$, report an occurrence at position $\mathrm{pos} + 1$.

(c) *Periodic shift.* Set $\mathrm{pos} \leftarrow \mathrm{pos} + p$, and $s \leftarrow m - p$ in the periodic case.

Example (Worked trace with memory). Let $P =$ `aabaabaab`. The maximal suffixes are `baabaab` for `a` $<$ `b` and P itself for `b` $<'$ `a`, so the cut is $\ell = 2$: $u =$ `aa`, $v =$ `baabaab` with $\mathrm{per}(v) = 3$, and since P has period 3 we are in the periodic case with $p = \mathrm{per}(P) = 3$. Search in $T =$ `xaabaabaabaaby`:



In iteration 1, the right phase sees $T[3] =$ `a` $\neq$ `b` $= P[3]$ and shifts by 1. In iteration 2, both phases match and the algorithm reports an occurrence. It then shifts by one period and sets $s = 6$: the first 6 characters are known to match. In iteration 3, the memory lets the right phase start at position 9 instead of 3, avoiding 6 redundant comparisons.

Theorem 7.2. The two-way algorithm finds all occurrences of P in T in $O(n + m)$ time using $O(1)$ extra space. The search phase performs fewer than $2n$ character comparisons.

Proof. Preprocessing takes $O(m)$ time and $O(1)$ space, as shown above.

We maintain the invariant that at the start of every iteration, $P[1..s] = T[\mathrm{pos} + 1\,..\,\mathrm{pos} + s]$. It holds initially with $s = 0$, and after a mismatch-driven shift or a shift in the non-periodic case, where $s = 0$. After a periodic shift in the periodic case, the right phase of the previous iteration gave $P[\ell{+}1..m] = T[\mathrm{pos}'{+}\ell{+}1..\mathrm{pos}'{+}m]$ for the old position pos', so at the new position $\mathrm{pos} = \mathrm{pos}' + p$ we have $T[\mathrm{pos}{+}1..\mathrm{pos}{+}m{-}p] = P[p{+}1..m]$, using $p \ge \ell + 1$, and $P[p{+}1..m] = P[1..m{-}p]$ by periodicity; so the invariant holds with $s = m - p$.

Every reported occurrence is genuine: the right phase verified $P[\ell{+}1..m]$, the left phase verified $P[s{+}1..\ell]$, and the invariant covers $P[1..s]$. No occurrence is skipped: a mismatch-driven shift by $i - \ell$ is justified by Lemma 7.2(i), where the partial match required by the lemma is provided by the invariant up to position s and by the scan beyond it; and a shift by p after the right phase matched in full is justified by Lemma 7.2(ii), since $p = \mathrm{per}(P)$ in the periodic case and $p \le \mathrm{per}(P)$ in the non-periodic case.

For the running time we count comparisons. Every right-phase comparison is made at a text position $\mathrm{pos}{+}i$ that strictly increases over the whole run: within a phase i increases; after a mismatch at i the next phase starts at $(\mathrm{pos}{+}i{-}\ell){+}(\ell{+}1) = \mathrm{pos}{+}i{+}1$; and after a full match, which ended at $\mathrm{pos}{+}m$, the next phase starts at $\mathrm{pos}{+}p{+}\max(\ell, s){+}1$, which is at least $\mathrm{pos}{+}m{+}1$ in the periodic case, because $p{+}\max(\ell, m{-}p) \ge m$, and at least $\mathrm{pos}{+}(m{-}\ell{+}1){+}\ell{+}1$ in the non-periodic case. So

there are at most n right-phase comparisons in total. Left-phase comparisons live inside the window $[\text{pos}+1, \text{pos}+\ell]$; since every left phase is followed by a shift of $p > \ell$, these windows are pairwise disjoint, contributing fewer than n comparisons in total. Adding the two bounds gives fewer than $2n$ comparisons in the search phase, and $O(n+m)$ time overall. □

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